



GE Medical Systems

Technical Publication

**Direction 2211226-100
Revision 3**

**GE Medical Systems
LightSpeed QXi InSite Manual**

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LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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このサービスマニュアルには英語版しかありません。

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このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

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注意:

本维修手册仅存有英文本。

非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (262) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

End of Section

Revision History

Revision	Date	Reason for change
0	5/18/1999	Initial release of Publication
1	11/19/1999	Graphical User Interfaces upadated to reflect current software. Appendix C changed to more accurately describe potential problems and their solution.
2	01/06/2000	CQA 998935. Modified Figure 3-14 to reflect new default switch settings for the USR Modem.
3	06/13/00	CQA 1001897 and 1001955. Added InSite CD-ROM Part and Version Numbers. Updated item 3 on page 49.

List of Effected Pages

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Preface

Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by its format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task. Please become familiar with the conventions used within this publication before proceeding.

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicated potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects



DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION



**WARNING
ROTATING
EQUIPMENT
BARE WIRES**

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.



**NOTICE
Equipment
Damage
Possible**

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0 Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant.

2.2 Page Layout

Publication Part Number & Revision Number

Publication Title

The current section and its title are always shown in the footer of the left (even) page.

An exclamation point in a triangle is used to indicate important information to the user.

Paragraphs preceded by **Alphanumeric** (e.g. numbers) characters is information that must be followed in a **specific order**.

The current chapter and its title are always shown in the footer of the right (odd) page.

Paragraphs preceded by **symbols** is (e.g. bullets) is information that has **no specific order**.

Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd

numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate directionality. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: Fixed Output This paragraph denotes computer screen fixed output. It's output is fixed from the sense that it does not vary from application to application. It's the most commonly used style used to indicate file-names, paths and text.

Example: Variable Output *This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>*

Example: Fixed Input **This paragraph denotes fixed input. It's typed input that will not vary from application to application. Fixed text the user is required to supply as input.**

Example: Variable Input ***This paragraph denotes computer input that can vary from application to application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.***

2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Soft Keys Whereas the computer **MENU** button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

End of Section

Chapter 1

InSite & IIP Introduction

Section 1.0 Overview:

This chapter gives a brief description of the Service and Customer Applications available with InSite and the InSite Interactive Platform.

Section 2.0 Service Features

2.1 InSite Class-M Software

The InSite software is not included in the System/Applications software set, but must be loaded from a separate Class-M CD-ROM. See [Chapter 2](#) for software loading instructions, and [Chapter 3](#) for software set-up and initialization procedures. The Class-M InSite CD-ROM that's appropriate for your system can be found in [Table 1-1 on page 21](#).

PROGRAM	H1.1 M4	H1.2 M3	H1.2 M4
CD Part No.	2243504-3	2243504-4	2243504-5
SW Version	6	7	8

Table 1-1 InSite CD-ROM Part and Version Numbers

2.2 InSite supports the following Service features:

- ProActive Diagnostics (ProDiags)
- Modem Configuration
- HealthPage
- InSite CheckOut

2.3 Service Applications Description

2.3.1 ProActive Diagnostics (ProDiags)

The InSite ProActive Diagnostics feature monitors diagnostically informative parameters of a scanner on a preset schedule. If the scanner has InSite remote dial-out capabilities, and is located in a Pole that has the networking infrastructure in place to allow contact, it can automatically dial the Automated Support Center (ASC) to report the test results which are then forwarded to Support Engineers at the respective Pole's OnLine Center. Some results will cause a dispatch to be created. Each scanner has its own test schedule during which testing can be performed. The field engineer should determine the normal working hours for the scanner, and when and whether ProActive Diagnostics should be run. A default schedule is supplied and should be fine for most systems. Most of the tests can run in the background while the operator is scanning. However, in the future, a test might require use of the scanner. While this type of intrusive test is running, the operator will be presented with a lockout screen. For emergency scanner use, the operator can interrupt intrusive testing by pressing the button that is displayed on the lockout screen.

Both the field engineer and the On-Line Center engineer can command ProDiags to perform these functions:

- Schedule tests
- Turn tests on and off
- Look at the log file
- Execute tests
- Look at test results

2.3.2 HealthPage

The HealthPage task is a ProDiags application that is designed to be used to provide quick access to logs and statistics that can be useful to the field engineer during planned maintenance and troubleshooting. The system health page contains data gathered over a specified time period. The On-Line Center will dispatch service if appropriate, and will retain some data for analysis and trending.

A system's health page report can be acquired from the system in one of four (4) ways:

- 1.) Automatically
- 2.) Remotely - via InSite
- 3.) On Site - via the Service Desktop
- 4.) By requesting it in an e-mail

2.3.3 Modem Configuration

Modem Configuration is a part of the InSite Interactive Platform Configuration tool. Modem Configuration allows the user to configure and set up the serial port and modem for use with InSite. The user must set (or accept the default configuration) for 6 items for the configuration to be applied. These items are:

- Dial-out Prefix
- Modem Type
- Serial Port Selection
- Dialing Mode
- Country
- Serial Port Speed

2.3.4 InSite Checkout

InSite Checkout is a part of the InSite Interactive Platform Configuration tool. InSite Checkout allows the user to complete the InSite configuration process. The user has one of three options available for performing checkout. These options are:

- Checkout Now
- Auto Checkout.
- Checkout Later

Section 3.0 Customer Features

3.1 InSite Interactive supports the following new customer features:

- InSite Interactive Platform
- InSite Interactive Home Page
- One Button Touch
- Customer Messages

InSite Interactive features are only available through the customer's purchase of an InSite Interactive license, that includes access to one or more of these features.

Note: The customer features presently are supported only in English.

3.2 Customer Applications Description

3.2.1 Product Invocation of InSite Interactive Applications

An additional icon/button will be added to the desktop environment allowing the operator easy access to the InSite Interactive Web based applications. In addition, the InSite icon will display 3 different pixmap images. These icons will be used to notify the operator of an InSite Event. The three different icons are displayed below (Figure 1-1).

Note: Logic to record InSite logons will still be available, the current Session Log application will remain unchanged.

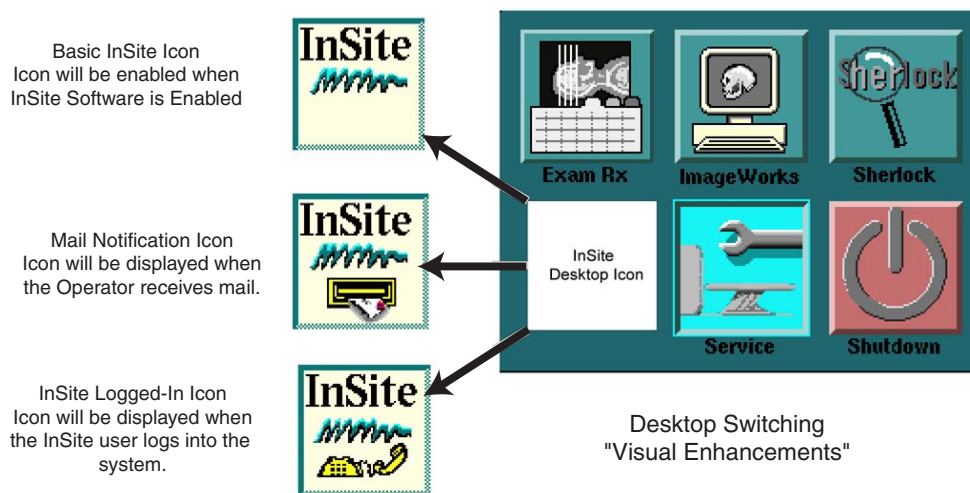


Figure 1-1 InSite Icons displayed to the operator.

3.2.2 System/Application Startup

The Application startup script will perform a check to determine if the InSite Interactive package contains a valid Service License (supplied by the OLC). If the check is successful, the Icon will appear on the desktop, otherwise the InSite icon will not be displayed, the Desktop will not be accessible and the Web Browser will not be started. In addition, during UNIX start-up, a script will be executed to see if the licenses are valid. If the licenses are valid, the Web Browser will start during UNIX initialization as a minimized application on the right video monitor. During Application Startup, the InSite Desktop Manager icon will be enabled and the Web Browser will be maximized

only within the InSite Desktop Environment. Regardless of the licenses' validation, the Web Server will always be enabled to allow InSite Web Server access and functionality.

3.2.3 Main InSite Interactive Application (InSite Home Page)

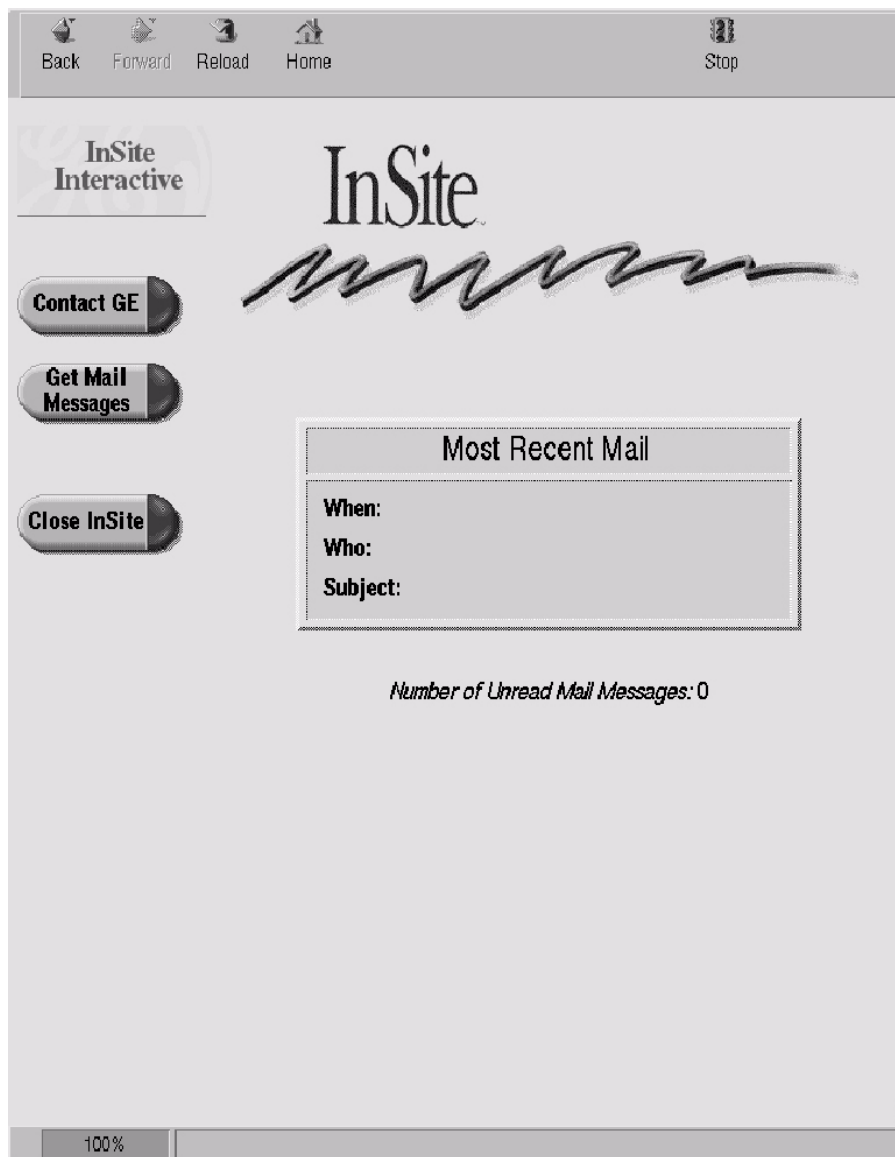


Figure 1-2 - InSite Interactive Home Page

This is the customers' top view of the InSite Interactive Service offerings. From this screen, the customer may access services which they are licensed to use. If the customer should select a button to a feature they do not have a license for, or have an expired license, they will be informed of this condition and instructed on how to receive a new license.

3.2.4 Contact GE Application / One Button Touch

The screenshot shows a web browser window with a navigation bar at the top containing icons for Back, Forward, Reload, Home, and Stop. The main content area has a header with the text "Contact GE" and a logo for InSite. Below the header, there is a section titled "Reason for Contacting GE:" with two buttons: "System Problem" and "Apps Question". To the left of the main content area is a vertical sidebar with three buttons: "Main Screen", "Contact GE", and "GE Phone Numbers". The "Contact GE" button is highlighted. Below the "Reason for Contacting GE:" section, there is a "Problem Area:" section with a list of checkboxes: Prescription, Archival, Image Quality, Acquisition, Filming, Other, Display, and Networking. Below this, there are input fields for "Submitter" (a dropdown menu labeled "select name") and "Other:". Below that, there are input fields for "Phone:" (a dropdown menu labeled "select phone number") and "Other:". Below that, there is a section for "Image (Exam/Series/Image) ==> E" with input fields for "S" and "I". Below this, there is a "Problem Description:" section with a large text area. Below the text area, there is a "Problem Date/Time:" section with a text field showing "4/6/1999 16:07". At the bottom right, there is a "Send to GE" button. The bottom status bar shows "100%".

Figure 1-3 - Contact GE Page.

The customer may enter a problem report and submit it to GE. The feature provides an easy way to capture important information which can result in faster repairs. The customer may also use this feature to ask a question of an application specialist. Status on the customer requests for assistance will be handled by a return message via the Customer Messages application.

3.2.5 Customer Messages / Customer Notify



Figure 1-4 - Customer Messages / Current Mail Page.

This feature permits GE to deliver a message to the customer on the product. The InSite Interactive feature will notify the customer that they have a new message by the button change mentioned in Section 3.2.1.

The customer may go to the Customer Messages area at any time by selecting the INSITE Button on the desktop, and then selecting Get Mail Messages from the Main InSite Interactive screen.

3.2.6 Additional InSite Interactive features

Additional features within the InSite Interactive application are being developed, and when they are available, will be able to be deployed by download to the system via the ASC. New features may require additional licenses to be purchased by the customer in order for those features to become available for their use.

Chapter 2

Class-M Installation Procedure

SOFTWARE INSTALLATION

This section describes the procedure to load Class-M software on a LightSpeed System from Class-M CD-ROM media.

- 1.) The system must have a modem and phone line installed prior to installing the Class-M software. Refer to the System Installation Manual for information on physical installation and configuration of the modem.
- 2.) Place the LightSpeed Class-M Software CD-ROM into the drive.

Note: You must use the latest version of the Class-M CD-ROM, [Table 1-1 on page 21](#), that is appropriate to the system software version. If you attempt to load an older version of Class-M software, you will be advised that you have an incorrect version of Class-M software, and you will be prevented from loading it.

- 3.) Open a UNIX shell on the system
Go to the Service Desktop and select UNIX SHELL from the tool chest.
- 4.) Change to root user. Type: `su -` ENTER
- 5.) Enter the root password.
- 6.) To start the installation, type: `install.classM` ENTER
- 7.) The software load should be complete when the script finishes, however, pay attention to any error messages that are generated during this process, as they may indicate unsuccessful loading of the InSite Software.
If everything completes normally, continue with InSite Configuration.

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Chapter 3

InSite Interactive Configuration

Section 1.0 Software Configuration

1.1 Overview Flowchart

This section provides instructions for installation/configuration of Class-M (IIP) software. The following flowchart is an overview of the procedure that is illustrated on the next several pages.

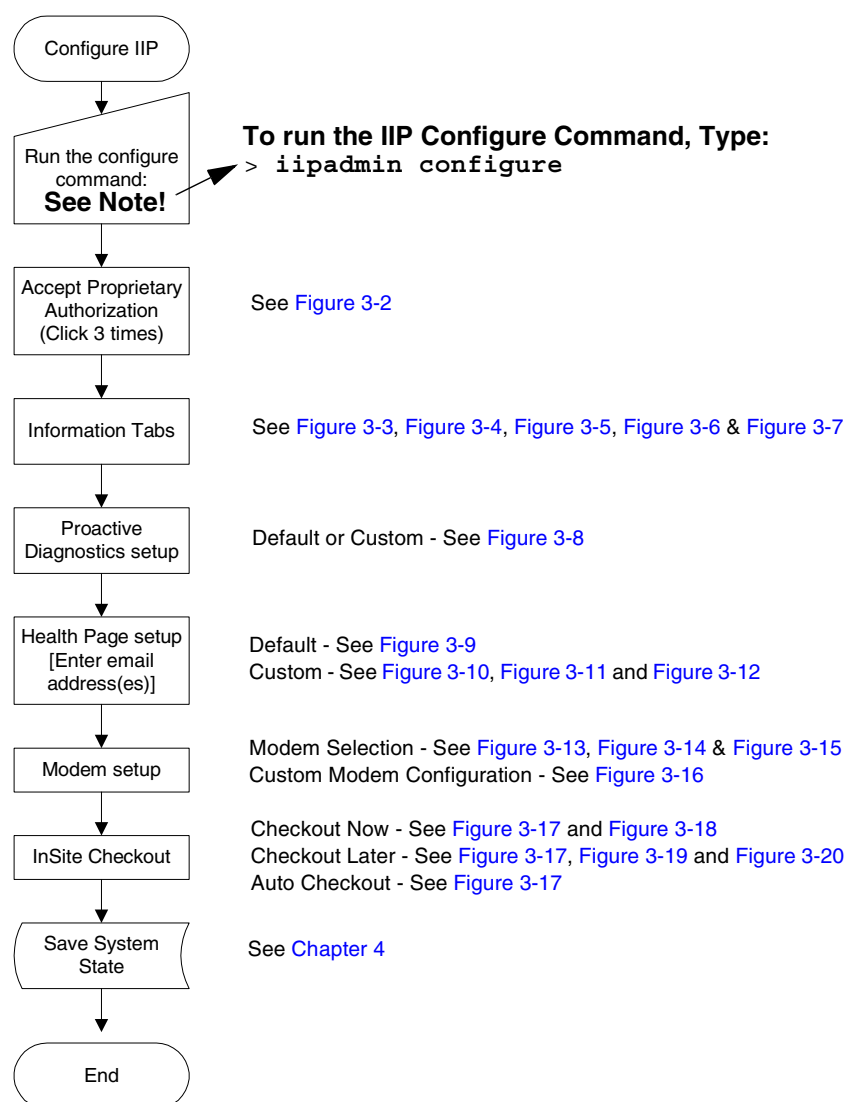


Figure 3-1 InSite Interactive (IIP) Configuration Flowchart

- 1.) Open a UNIX SHELL from the SERVICE DESKTOP.
- 2.) Change to root user. Type: `su -` **ENTER**

- 3.) Enter the root password.
- 4.) To start the configuration process, type: `iipadmin config` ENTER

1.2 Configuration Description

Note: The following pages provide screen snapshots and descriptions of the IIP Configuration Tool that starts up when the user executes the command “`iipadmin config`”.

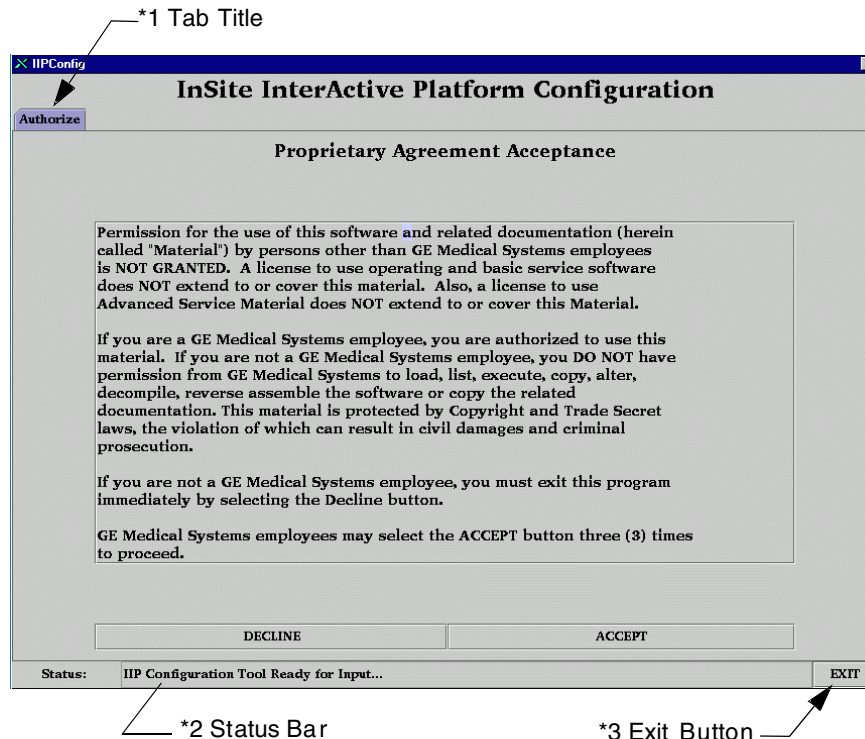


Figure 3-2 Proprietary Authorization Tab

The Authorization Tab, [Figure 3-2](#), is a security confirmation screen. The user is prompted to click on DECLINE if not a GE employee or click on ACCEPT 3 times if the user is a GE employee. Basic components of the window are:

- *1 - all tabs contain a title,
- *2 - the status area that displays status of actions, and
- *3 - an Exit button to exit the IIP Configuration Tool.

After ACCEPT has been clicked on three times, the main tabs are initialized.

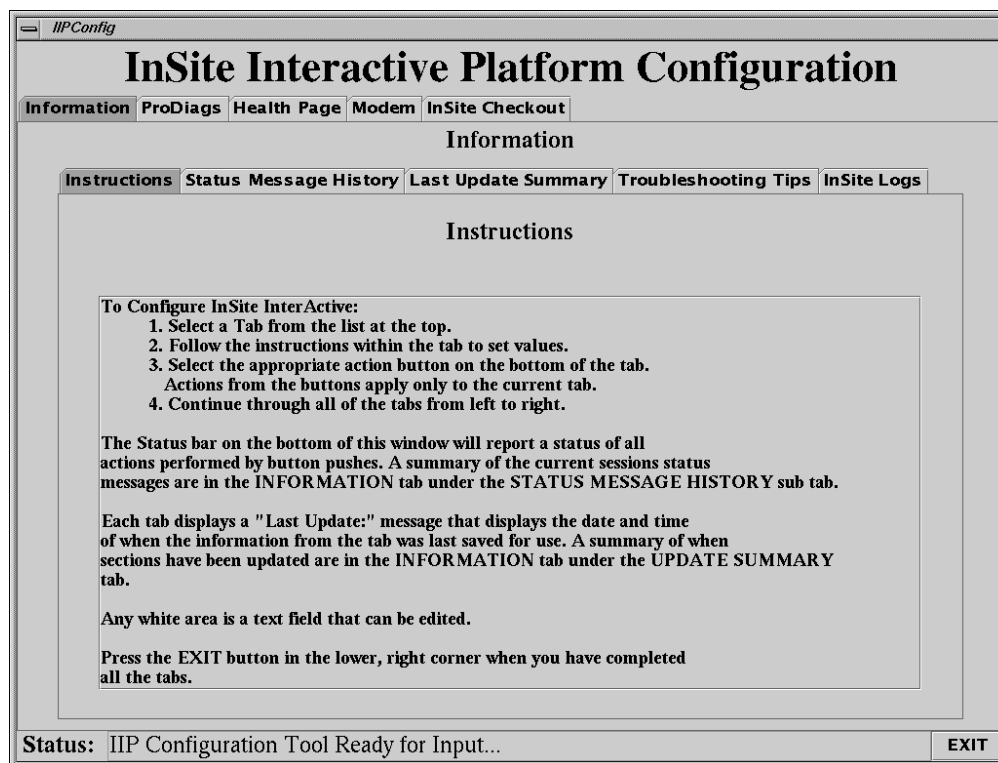


Figure 3-3 Information Tab, Instructions sub-tab

The Information tab contains 4 sub-tabs. The Instructions tab, [Figure 3-3](#), provides the user with information on how to use the IIP Configuration Tool. Briefly, to configure InSite the user is to select the tabs from left to right. Instructions in each of the tabs will assist the user in executing the tasks on that tab. When all tabs are completed, the user may exit the tool.

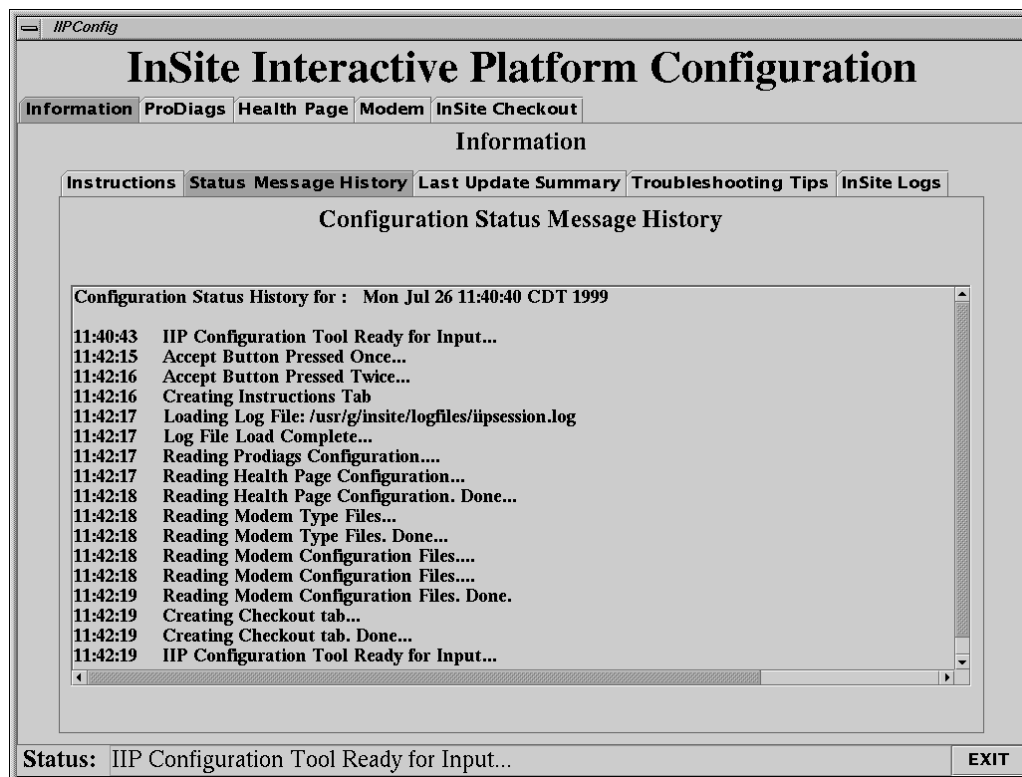


Figure 3-4 Information Tab, Status Message History sub-tab

The Information Tab, Status Message History sub-tab, [Figure 3-4](#), provides the user an up-to-date listing of all the status messages that have appeared in the Status Message Line at the bottom of the Tool window. Text displayed in this window is for the current session only. All status messages are tagged with a time only stamp. When the IIP Configuration Tool is run using the -debug switch, additional messages are displayed here as well as logged to the file `/usr/g/insite/logfiles/configdebug.log`.

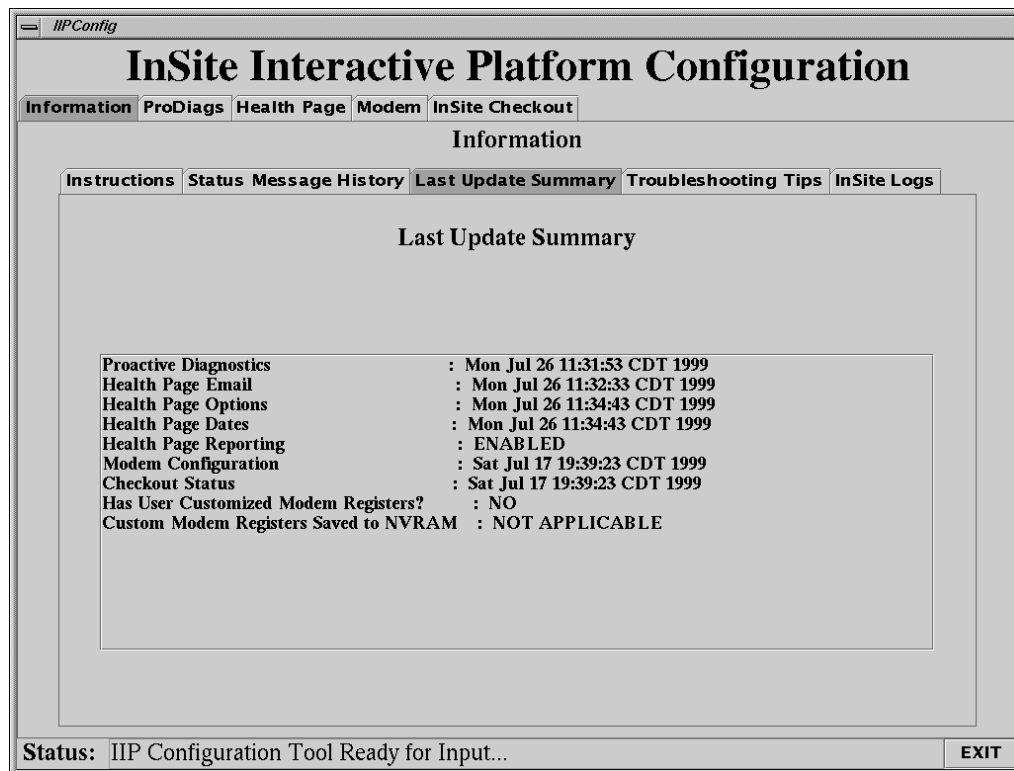


Figure 3-5 Information Tab, Update Status List sub-tab

The Information Tab, Update Status List sub-tab, [Figure 3-5](#), provides the user with information on when each package was last configured.

On this page, the indication "NO FILE" means that the particular option has not been configured yet.

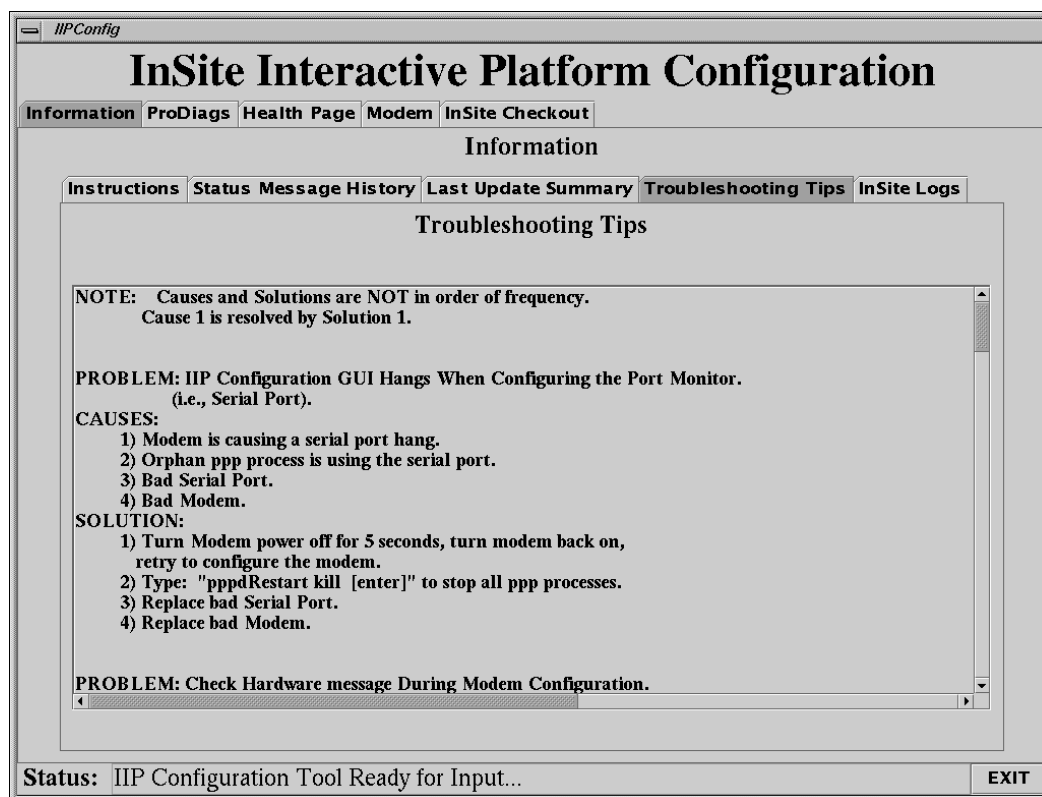


Figure 3-6 Information tab, Troubleshooting Tips sub-tab

The Information tab, Troubleshooting Tips sub-tab, [Figure 3-6](#), provides the user with basic Problem/Cause/Solution for InSite related problems.

This same problem/solution information text is contained in this document in [Appendix C](#), and is in the form of Troubleshooting Flowcharts in [Chapter 6](#).

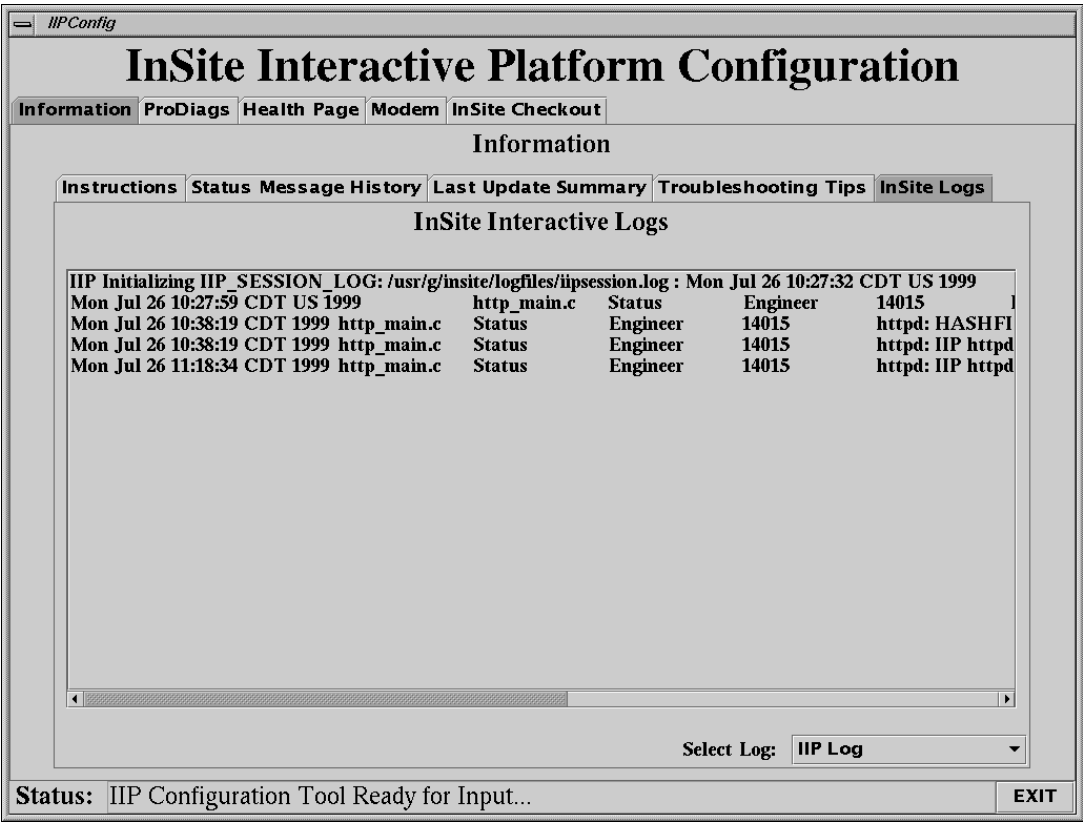


Figure 3-7 Information tab, InSite Logs sub-tab

The Information tab, InSite Logs sub-tab, see [Figure 3-7](#), provides the user with the ability to view the various IIP-related log files. (Use the `Select Log`: pull-down menu.)

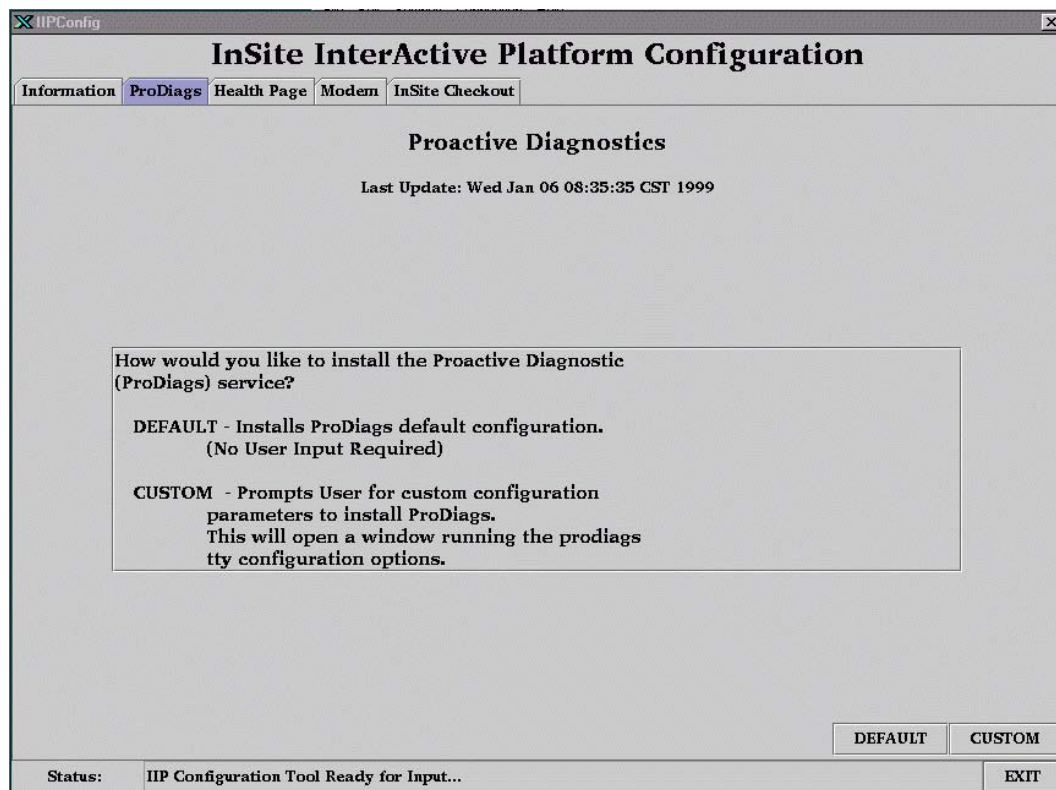


Figure 3-8 Proactive Diagnostics Tab

The ProDiags Tab, [Figure 3-8](#), is for the configuration of Proactive Diagnostics. The user is to select either the DEFAULT button or the CUSTOM button to configure ProDiags.

Selecting the DEFAULT Button will configure ProDiags with the schedule files provided.

Selecting the CUSTOM button allows the user to modify the default ProDiags schedule for the site.

InSite Interactive Platform Configuration

Information ProDiags **Health Page** Modem InSite Checkout

Health Page

Last Update: Mon Jul 26 11:32:33 CDT 1999

Enter or edit E-mail addresses to receive the Health Page report.
You MUST enter at least one address.

NOTE: Include the full Internet address or you will not receive the report!
Example: FirstName.LastName@med.ge.com

DEFAULT – Updates the Health Page E-mail addresses only.

CUSTOM – To gain access to Custom Health Page settings.
NOTE: E-mail addresses MUST be entered before entering Custom Health Page

☒ HealthPage Report Enabled

Enter Address List

- tzortzos@ct.med.ge.com
- byron.matthews@med.ge.com
- fragosso@ct.med.ge.com
- getchel@ct.med.ge.com

DEFAULT CUSTOM >>

Status: IIP Configuration Tool Ready for Input... EXIT

Figure 3-9 Health Page Tab

The Health Page Tab, [Figure 3-9](#), is for the configuration of Health Page. The user is to select either the DEFAULT or the CUSTOM button to configure Health Page.

Selecting the DEFAULT Button will configure Health Page with the a default schedule file. A valid e-mail address is required in the Enter Address List box for the DEFAULT configuration. A valid e-mail address is defined as a string in the format `<name>@<location>`.

Remember that the standard format for GEMS e-mail addresses is in the form:

`<firstname>.<lastname>@med.ge.com`. Be careful to enter a correct e-mail address(es).

A check is made for an “@” only. All e-mail addresses are saved to the

`/usr/g/config/healthpg.adr` file in the format: `user<#>=<email address>`

Note: You must ensure that the Health Page Report Enabled checkbox is checked to enable creation of the Health Page Report.

Selecting the CUSTOM button writes the e-mail address file and generates the Custom Health Page sub-tabs. If you have not previously changed the healthpage schedule, a list of pm dates is created starting one month from the current date.

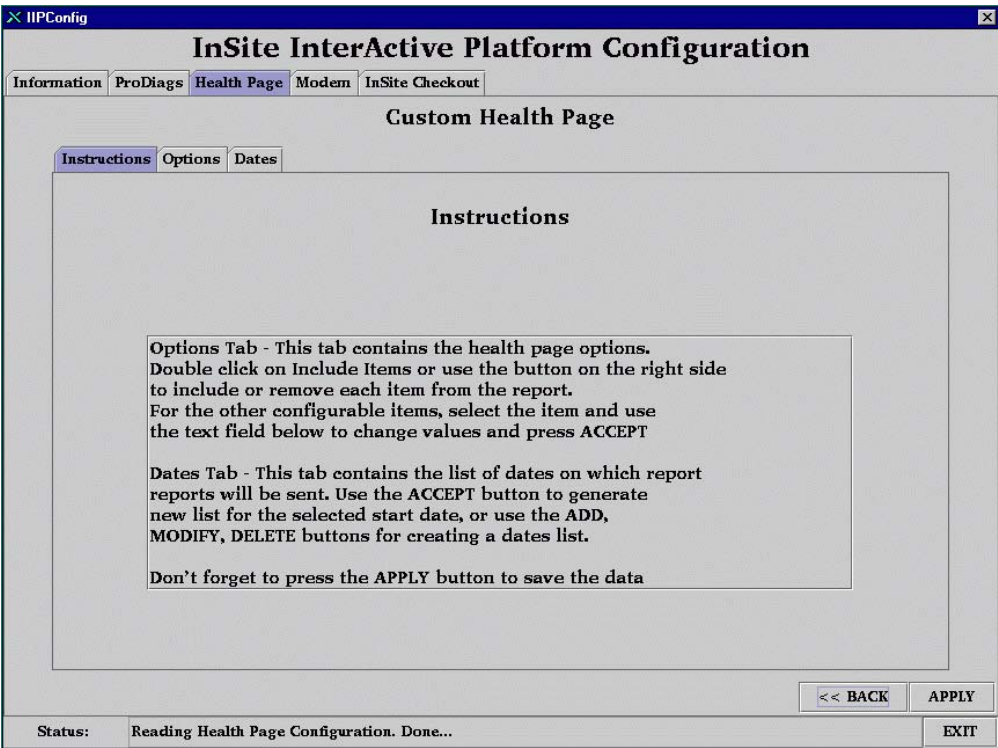


Figure 3-10 Custom Health Page Instructions Tab

The Custom Health Page Tab, Instructions sub-tab, [Figure 3-10](#), provides the user with instructions on changing values in the Options and Dates sub-tabs.

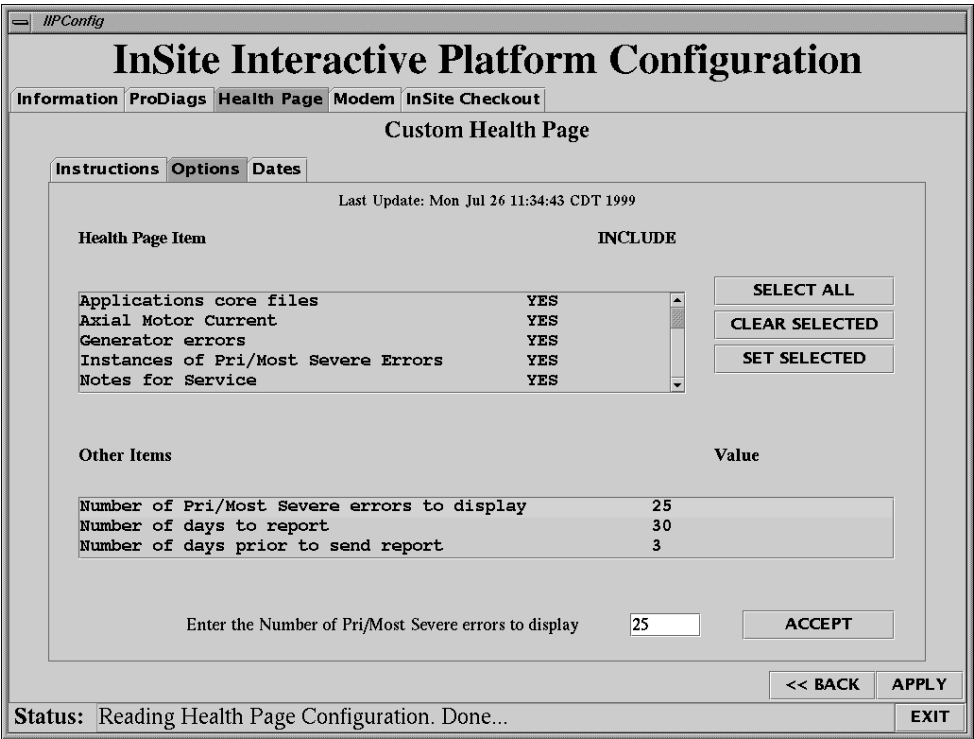


Figure 3-11 Custom Health Page Options Tab

In the Custom Health Page Options Tab, [Figure 3-11](#), the user can configure Health Page contents. Items in the Health Page Item box can be changed by double clicking on the item to toggle from YES to NO or NO to YES. The SELECT ALL button will select all of the items. The CLEAR SELECTED button will set all selected items to NO. The SET SELECTED button will set all selected items to YES.

The OTHER ITEMS box contains items which are value based. Click on the item to select, then edit the value in the box below. Select the ACCEPT box to change the value in the list. Selecting the APPLY button will write the values to the /usr/g/config/healthpg.cfg file

The screenshot displays the 'InSite InterActive Platform Configuration' window. The 'Health Page' tab is selected, and within it, the 'Dates' sub-tab is active. A list of dates is shown under 'Health Page Dates', ranging from 05/15/1997 to 03/11/1998. To the right, there are two date input sections: 'Enter Start Date' and 'Enter Specific Date'. The 'Enter Start Date' section has dropdowns for 'FEB', '1', and '1999', with an 'Update List' button below. The 'Enter Specific Date' section has dropdowns for 'MAY', '15', and '1997', with 'ADD', 'MODIFY', and 'DELETE' buttons below. At the bottom right are '<< BACK', 'APPLY', and 'EXIT' buttons. The status bar at the very bottom indicates 'Status: Reading Health Page Configuration. Done...'.

Figure 3-12 Custom Health Page Dates Tab

In the Custom Health Page Dates Tab, [Figure 3-12](#), the user can configure Health Page pm dates. Health page reports will be generated several days prior to a pm date and e-mailed to the FE.

The ENTER START DATE area generates a new pm date list based on the new date selected from the pull down menus. The UPDATE LIST button will generate the new list and put it in the Health Page Dates list box.

By selecting a date in the Health Page List box, the user may modify or delete the date by selecting the MODIFY or DELETE buttons. The selected date appears in the ENTER SPECIFIC DATE pull downs. The ENTER SPECIFIC DATE pull downs may also be used to enter a single date into the list by changing the date in the pull downs to the desired date, then selecting the ADD button.

The APPLY button must be selected to update the changed dates into the /usr/g/config/healthpg.pm file.

The screenshot shows a web-based configuration interface titled "InSite InterActive Platform Configuration". At the top, there are tabs for "Information", "ProDiags", "Health Page", "Modem", and "InSite Checkout", with "Modem" currently selected. Below the tabs, the heading "Modem Configuration" is centered, followed by the text "Last Update: Tue Apr 06 15:28:48 CDT 1999". A box contains instructions: "Select the Modem settings given below and Press APPLY - To update the modem and modem configuration files with model and country defaults. CUSTOM - For other modem settings." Below this, there are six configuration fields arranged in two columns: "Dial-out Prefix" (set to "9,1"), "Dialing mode" (set to "Tone"), "Modem Type" (set to "USRobotics Courier V.34 or V.Everything"), "Country" (set to "Default - All Others"), "CPU Serial Port Name" (set to "/dev/ttyf1"), and "CPU Serial Port Speed" (set to "38400"). At the bottom right of the form are buttons for "APPLY" and "CUSTOM >>". A status bar at the very bottom shows "Status: Health Page Email address file Update: Completed Successfully" and an "EXIT" button.

Figure 3-13 Modem Configuration Tab

The Modem Tab, [Figure 3-13](#), allows the user to configure and set up the serial port and modem for use with InSite. The user must set 6 items for configuration to be applied. The items are Dial-out Prefix, Dialing Mode, Modem Type, Country, Serial Port Selection, and Serial Port Speed.

- 1.) The default settings for CPU Serial Port Name and CPU Serial Port Speed should be correct, and should not be changed unless troubleshooting.
- 2.) The user must either select a Dial-out prefix or enter a site specific one if it is not in the list. Dial-out prefixes may be required on some sites to get a line outside of the hospital. "9,1" is the most common and is the default.
- 3.) The user must select a dialing mode of Tone or Pulse. Tone is selected by default.
- 4.) The user must select one of the supported modem types. See [section Table 3-1 on page 41](#) for a cross reference of modems listed in CT versions of InSite prior to IIP, and the IIP modem list.
- 5.) The user must set the Country in which the site is located. "Default - All Others" is selected by Default. If your Country is not listed in the pull-down menu, then the correct selection is "Default - All Others".

After all selections are made the user is to select the APPLY button to set up the serial port and set up the registers on the modem, and store the values to the modem's NVRAM. When the APPLY button is pressed, another screen will be displayed for the selected modem type (see [Figure 3-14](#)).

Note: You **MUST** press the APPLY button before continuing on to either the InSite Checkout Tab or the Custom Modem Configuration screen.

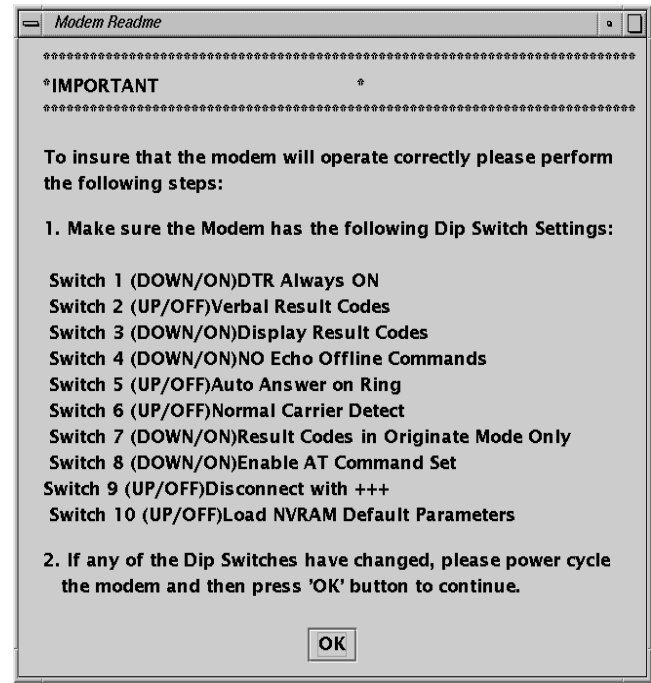


Figure 3-14 Modem "Readme" screen (example is for USB V.34 selection)

This screen is to remind the user to properly configure any switches or other modem parameters at the modem, to prepare the modem to accept the programming. Once the OK is selected on this screen, the program will proceed to programming of the modem, and will return with the screen display of [Figure 3-15](#) (similar to [Figure 3-13](#)).

IIP MODEM LIST
Hayes Optima V.FC and V.34
Motorola Codex 3260 Fast V.34
USRobotics Courier V.34 or V.Everything
Multitech MT56342BA V.90
3Com Office Connect 56K Business Modem

Table 3-1 Modem cross-reference table

The screenshot shows a web-based configuration interface titled "InSite InterActive Platform Configuration". At the top, there are navigation tabs: "Information", "ProDiags", "Health Page", "Modem" (which is selected), and "InSite Checkout". Below the tabs, the main heading is "Modem Configuration". Underneath this heading, it says "Last Update: Tue Apr 06 15:28:48 CDT 1999". A text box contains instructions: "Select the Modem settings given below and Press". Below this, there are two options: "APPLY - To update the modem and modem configuration files with model and country defaults." and "CUSTOM - For other modem settings." Below these instructions, there are several configuration fields: "Dial-out Prefix" with a dropdown menu showing "9,1"; "Dialing mode" with a dropdown menu showing "Tone"; "Modem Type" with a dropdown menu showing "USRobotics Courier V.34 or V.Everything"; "Country" with a dropdown menu showing "Default - All Others"; "CPU Serial Port Name" with a dropdown menu showing "/dev/ttyf1"; and "CPU Serial Port Speed" with a dropdown menu showing "38400". At the bottom right of the configuration area, there are two buttons: "APPLY" and "CUSTOM >>". At the very bottom of the window, there is a status bar that says "Status: Health Page Email address file Update: Completed Successfully" and an "EXIT" button.

IIIPConfig

InSite InterActive Platform Configuration

Information ProDiags Health Page **Modem** InSite Checkout

Modem Configuration

Last Update: Tue Apr 06 15:28:48 CDT 1999

Select the Modem settings given below and Press

APPLY - To update the modem and modem configuration files with model and country defaults.

CUSTOM - For other modem settings.

Dial-out Prefix
9,1

Dialing mode
Tone

Modem Type
USRobotics Courier V.34 or V.Everything

Country
Default - All Others

CPU Serial Port Name
/dev/ttyf1

CPU Serial Port Speed
38400

APPLY CUSTOM >>

Status: Health Page Email address file Update: Completed Successfully EXIT

Figure 3-15 Updated Modem Configuration Tab

The CUSTOM button is used to display a screen of custom modem selections that are not normally required in a standard setup. If your site/country setup requires Custom Modem settings, please contact your local On-Line Center for details, and refer to [Figure 3-16](#).

InSite InterActive Platform Configuration

Information ProDiags Health Page **Modem** InSite Checkout

Custom Modem Configuration

Modified Registers: NO Modified Registers Saved to NVRAM: NOT APPLICA...

WARNING: CHANGING SETTINGS ON THIS SCREEN MAY DISABLE INSITE. USE WITH CAUTION ONLY TO RESOLVE A SPECIFIC CONNECTION ISSUE. REPORT CHANGES BACK TO THE INSITE DEVELOPMENT TEAM FOR FUTURE UPDA

EXECUTE – Press this button to execute the Action selected

USRobotics Courier V.34 or V.Everything

Register S0 Register Value 0

Register Description Auto Answer Ring Count

Register String Saved to NVRAM &F1 X4 &D0 E0 T M0 Q2 S0=1 S7=90

Action Update register

<< BACK EXECUTE

Status: EXIT

Figure 3-16 Custom Modem Configuration screen (example is for USR V.34 selection)

If you need to make any modifications to this screen (such as making specific changes to a modem register or other modem parameter, as directed by your local On-Line Center), you **MUST** also change the ACTION pull-down to read WRITE TO NVRAM. Then when you press the EXECUTE button, the changed values will be applied both to the modem's registers, and also will be stored in the modem's NVRAM. This will ensure that the changed modem settings will be saved and restored whenever the modem's power is cycled.

To exit this screen, and return to the previous Modem Configuration Tab screen (Figure 3-15), press the BACK button.

The Custom Modem Tab, Figure 3-16, allows the user to modify the set up registers on the modem. Extreme caution should be used when changing anything on this tab. InSite may be disabled if improper settings are sent to the modem.

The Register and Register Value fields work as a pair. A user may select an "s" register to change a value or enter the name of a non-S register to be queried or modified. The REGISTER VALUE field is where to enter the new value of the register the user wants to change. If an "s" register is selected from the pull down menu, a description of that register will be displayed in the REGISTER DESCRIPTION.

REGISTER STRING is a list of registers and values set in the modem by default, based on the Modem Type and Country selected in the Modem Tab. This string updates when a register has been modified and the settings have been saved to NVRAM from the Custom Modem Tab.

The ACTION pull down menu provides 6 actions to be executed by the EXECUTE button. The actions are UPDATE REGISTER, READ REGISTER, DISPLAY ALL, WRITE TO NVRAM, FACTORY DEFAULTS, and CHECK MODEM.

- UPDATE REGISTER tries to set the Register defined in the Register field to the value defined in the Register Value field.
- READ REGISTER takes the Register defined in the Register field and returns the value of the register in the status bar.
- DISPLAY ALL opens a window with all the register settings.
- WRITE TO NVRAM forces the modem to take the temporary modem settings and save them to NVRAM within the modem. These values in NVRAM are recalled when the modem's power is cycled. **This is the action that MUST be selected when changing modem settings, or the next time power is cycled to the modem, it will revert to its factory default original settings.**
- FACTORY DEFAULTS resets the modem to factory defaults using a hardware flow control template.
- CHECK MODEM is a simple check to see if commands can be sent to the modem.

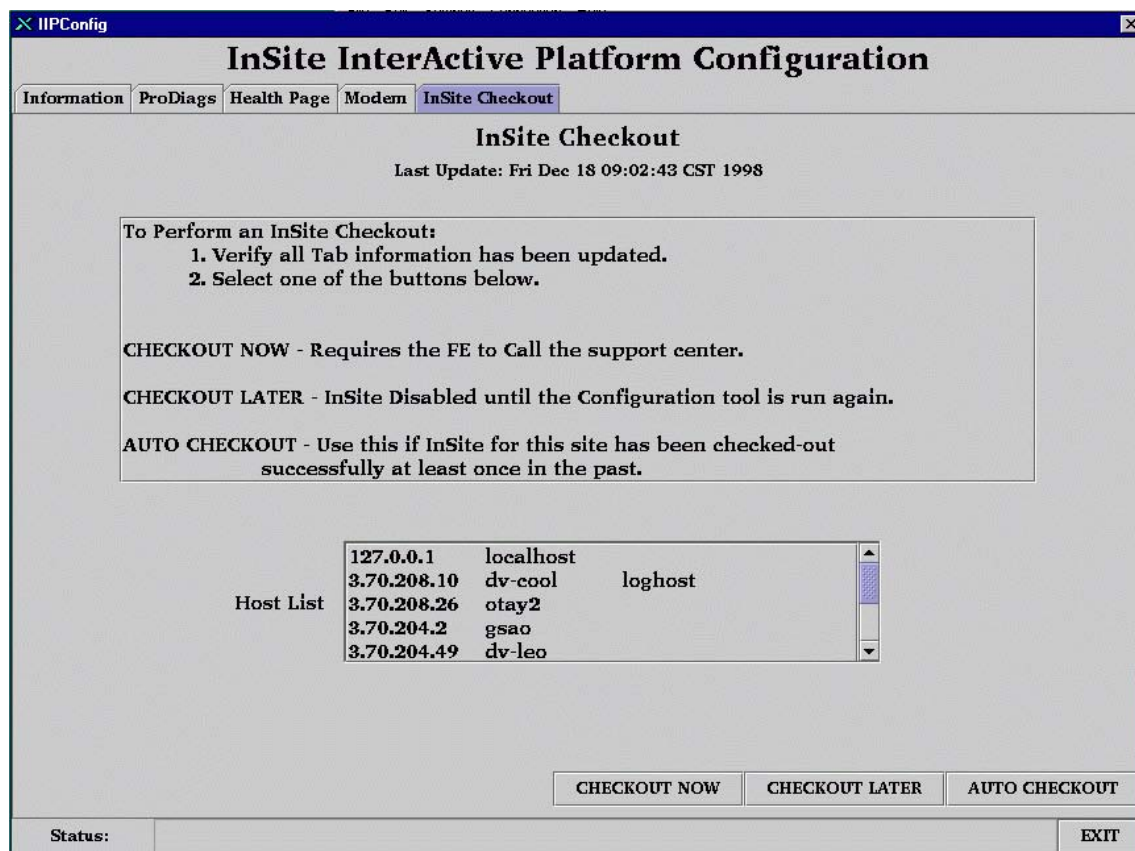


Figure 3-17 InSite Checkout Tab

The InSite Checkout Tab, [Figure 3-17](#), allows the user to complete the InSite configuration process. The user has one of three options, CHECKOUT NOW, CHECKOUT LATER, or AUTO CHECKOUT.

- Selecting the CHECKOUT NOW button configures ppp to allow a dial in connection from the OnLine Center. The user must call the local OnLine Center to have complete the checkout. The user will be asked to provide information from the "host list" section of the figure above (this is a listing of all host entries in the /etc/hosts file on the system). Locate the "loghost" entry, and check that the IP Address listed is correct prior to attempting the Checkout. During the Checkout process, the OnLine Center will download a sclink.cfg configuration file and run ConfigLink to configure ppp to allow only the OnLine Center to be able to dial and login to the site. See [section Figure 3-18 on page 46](#) for more information.
- Selecting the CHECKOUT LATER button will disable the serial port for the modem for a checkout at a later date. All previously saved information is retained. If this is selected the user MUST rerun the IIP Configuration Tool, select the APPLY button on the Modem Tab, and then select the CHECKOUT NOW button. See [section Figure 3-19 on page 46](#) for more information.
- The AUTO CHECKOUT button can only be used if a sclink.cfg file exists, (i.e., meaning the site has completed a checkout using the CHECKOUT NOW button at some previous time. AUTO CHECKOUT will run ConfigLink, dial the AutoSC, queue a task with the AutoSC to dial back, and wait for the AutoSC to dial back and leave a file /tmp/autocheckout. The IIP Configuration Tool will wait only 10 minutes for the file to appear. Typically, the AutoSC will call back in 5-6 minutes.

Note: When the dialout test is in the process of connecting to the AutoSC, the IIP Configuration GUI will not update and may appear locked up until the connection to the AutoSC is completed or the connection times out.

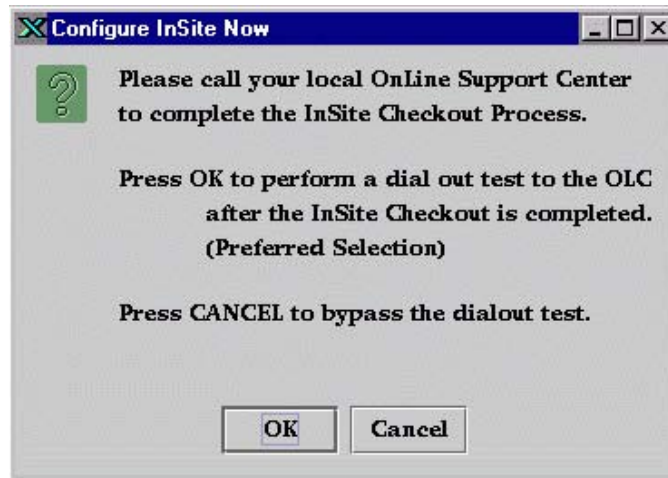


Figure 3-18 Configure InSite Now Requester

When the user selects the CHECKOUT NOW button, the CONFIGURE INSITE NOW requester, [Figure 3-18](#), is displayed providing the user the option to run a dial-out test upon completion of the InSite Checkout Process. If the user selects the OK button before the InSite Checkout Process is complete, the IIP Configuration Tool watches for a `~insite/sclink.cfg` file to appear. The IIP Configuration Tool will only wait 10 minutes for the `sclink.cfg` file to arrive. When the file appears, the dial-out test starts. This test ensures the dial prefix provided by the user and the number of the OnLine Center are correct. This also ensures Proactive Diagnostics messages and Contact GE requests can be sent to the OnLine Center.



Figure 3-19 Configure InSite Later Requester

When the user selects the CHECKOUT LATER button, the CONFIGURE INSITE LATER requester, [Figure 3-19](#), is displayed. The serial port is disabled and the IIP Configuration Tool will exit after the user selects the OK button.

Remember that you **MUST** rerun the IIP Configuration Tool at a later date, select the APPLY button on the Modem Tab, and then select the CHECKOUT NOW button to complete the InSite configuration.

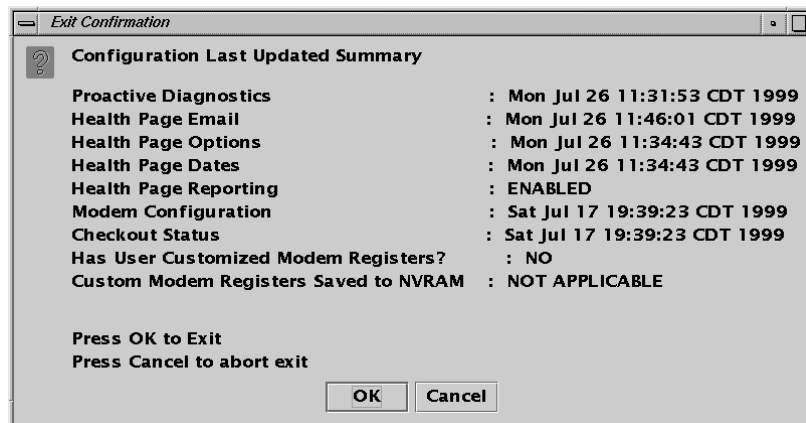


Figure 3-20 Exit Confirmation Requester

The EXIT CONFIRMATION REQUESTER, [Figure 3-20](#), is displayed when the user presses the EXIT button in the lower right corner of the IIP Configuration Tool Window.

A Last Updated summary is provided to the user. At this point, if you configured all InSite functions, a valid date/time should be displayed for each function. If any function shows "NOT CONFIGURED", that function was not properly configured.

The user can select the OK button to complete the exiting of the IIP Configuration Tool or select CANCEL to return to the IIP Configuration Tool.

Chapter 4

Saving State of the InSite Configuration

SAVE SYSTEM STATE

The configuration files for Class-M software and InSite features are saved to MOD as part of the Save System State process.

- 1.) Since the files which are necessary to back up a system prior to update or reload are dynamic, InSite Interactive can be backed up by performing System State Save, and ensuring that the Reconfig Info button is selected in the list of items to save.
- 2.) The type of files that will be saved are: the license files, the setup files for the modem connectivity, the configuration settings for the web server, the customer messages files, reports, and the identification of downloaded software packages.
- 3.) Any updated or downloaded software packages will not be included in the backup. In order to restore these downloaded files, an update request must be made to the OLC following a restore. This is done by contacting the OLC to request the download of any patches that apply to the current revision of software on your system and/or running the Auto InSite Checkout following a LFC.

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Chapter 5

De-Installation of the Class-M Software

SOFTWARE REMOVAL

This section describes the procedure to remove and disable Class-M software on a LightSpeed System.

- 1.) To remove the Class-M Software applications from the system, perform the following commands:
- 2.) Open a UNIX shell on the system:
Go to the Service Desktop and select UNIX SHELL from the tool chest.
- 3.) Change to root user. Type: `su -` - ENTER
- 4.) Enter the root password.
- 5.) Type: `uninstall.classM` ENTER
- 6.) This will remove the Class-M software package and disable InSite.
- 7.) Once you have removed Class M Software applications from the system, you must re-save your System State (ensure that Reconfig Info is selected).

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Chapter 6

InSite Interactive Service Information & Troubleshooting

Section 1.0 Troubleshooting Overview

If you should encounter any problems while installing the Class-M software package, or while configuring InSite Interactive, please refer to the flow-charts and additional information in this section to determine and correct the cause of your problem.

The following table shows the correlation between IIP defined directories, parameters, and files, as described in the troubleshooting flowcharts on the next pages, and the actual directories, parameters, or file locations on the system.

IIP DEFINED VALUE	SYSTEM DEFINITION
\$INSITE_HOME	/usr/g/insite
\$IIP_MODALITY_INFO	/usr/g/config/INFO
\$IIP_CONFIG_DIR	/usr/g/config
\$IIP_HEALTHPAGE_DIR	/usr/g/bin
\$IIP_HP_CONFIG_DIR	/usr/g/config
\$IIP_II_MODALITY	CT
\$IIP_SESSION_LOG	/usr/g/insite/logfiles/iipsession.log
\$IIP_VIEWLOG_FILENAME	/usr/g/service/log/gesys_{hostname}.log
\$IIP_NOTIFY_II_LOGIN	/usr/g/bin/ctiiputil -login
\$IIP_NOTIFY_II_LOGOUT	/usr/g/bin/ctiiputil /logout

Table 6-1 IIP Configuration Parameter Settings

Section 2.0 Troubleshooting Flowcharts

2.1 Configuration fails

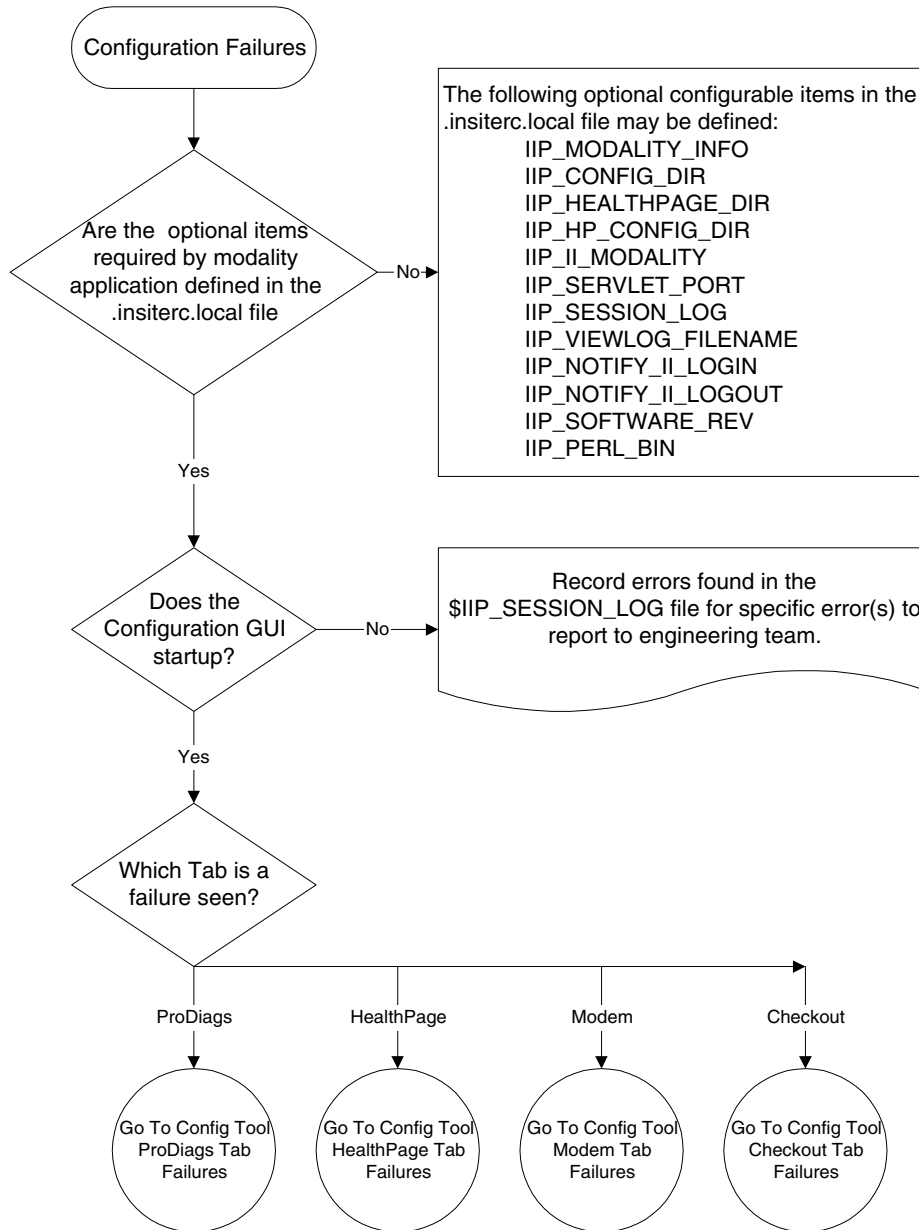


Figure 6-1 Configuration Fails - Flowchart

2.2 Config Tool ProDiags Tab Failures

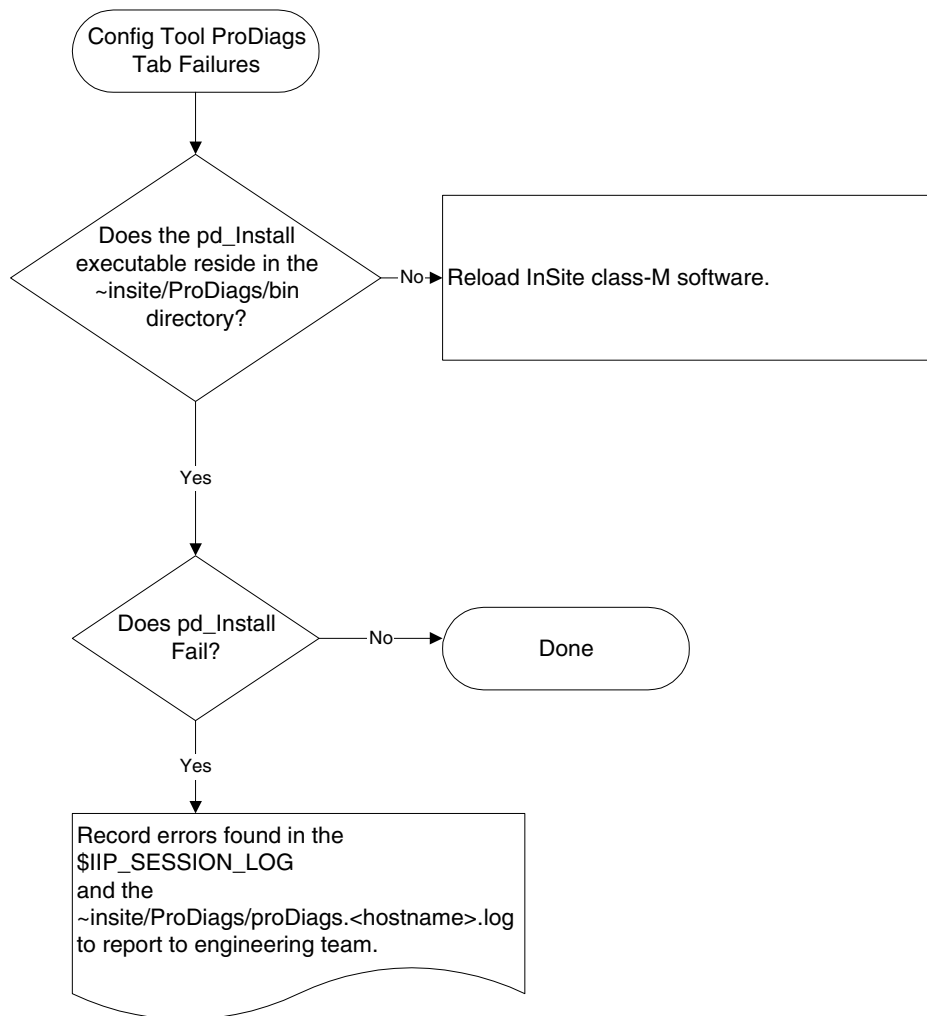


Figure 6-2 Config Tool ProDiags Tab Failures - Flowchart

2.3 Config Tool Health Page Tab Failures

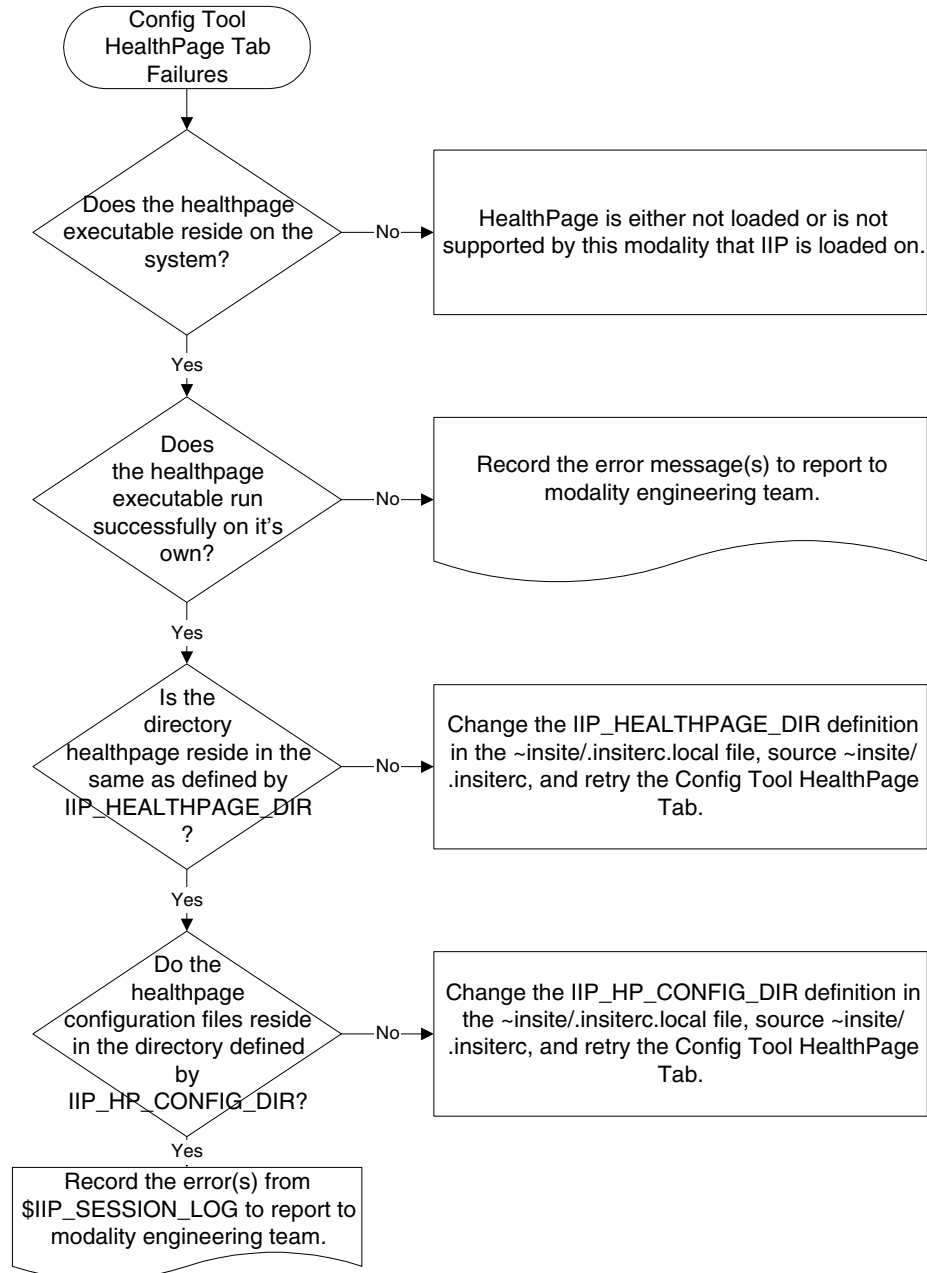


Figure 6-3 Config Tool Healthpage Tab Failures - Flowchart

2.4 Config Tool Modem Tab Failures

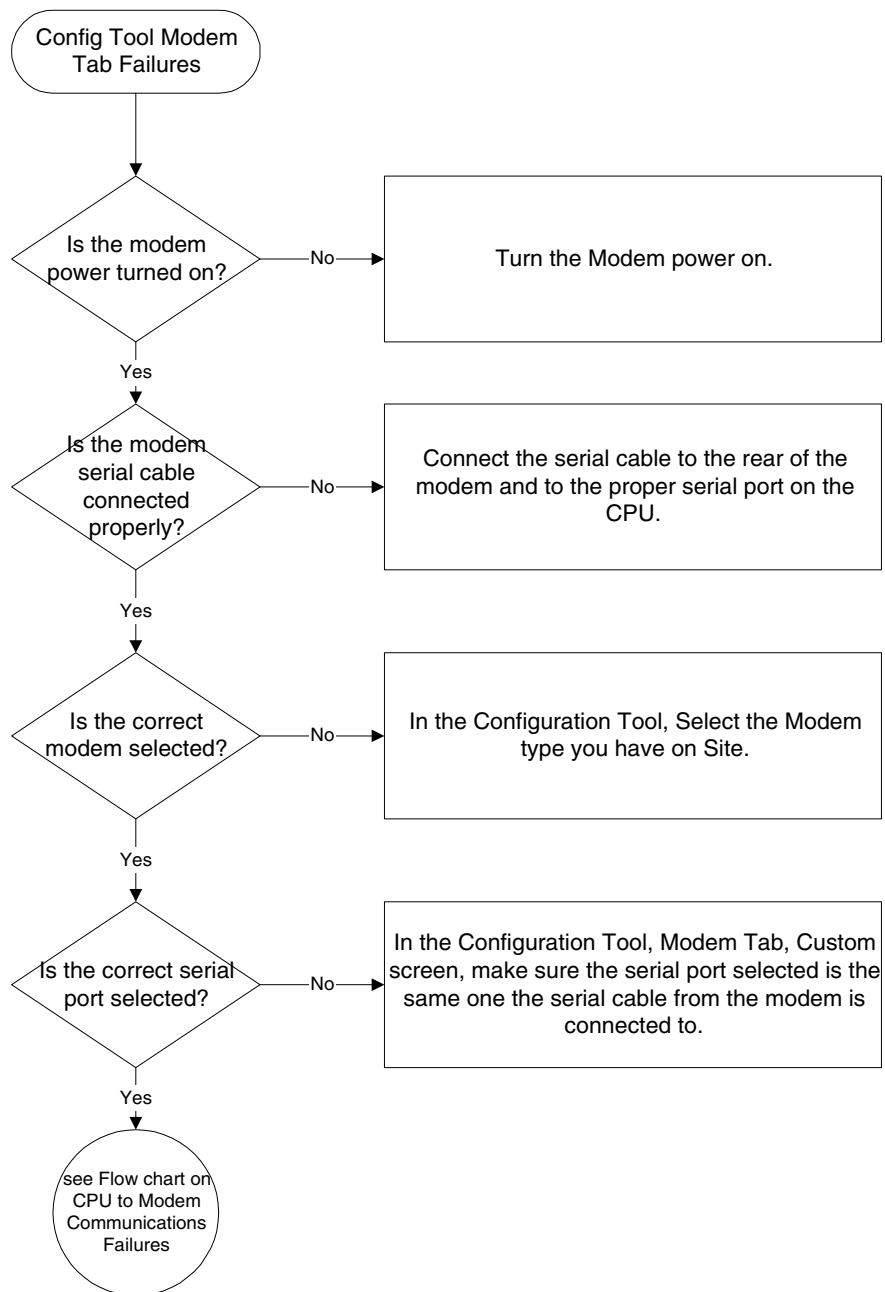


Figure 6-4 Config Tool Modem Tab Failures - Flowchart

2.5 Config Tool Checkout Tab Failures

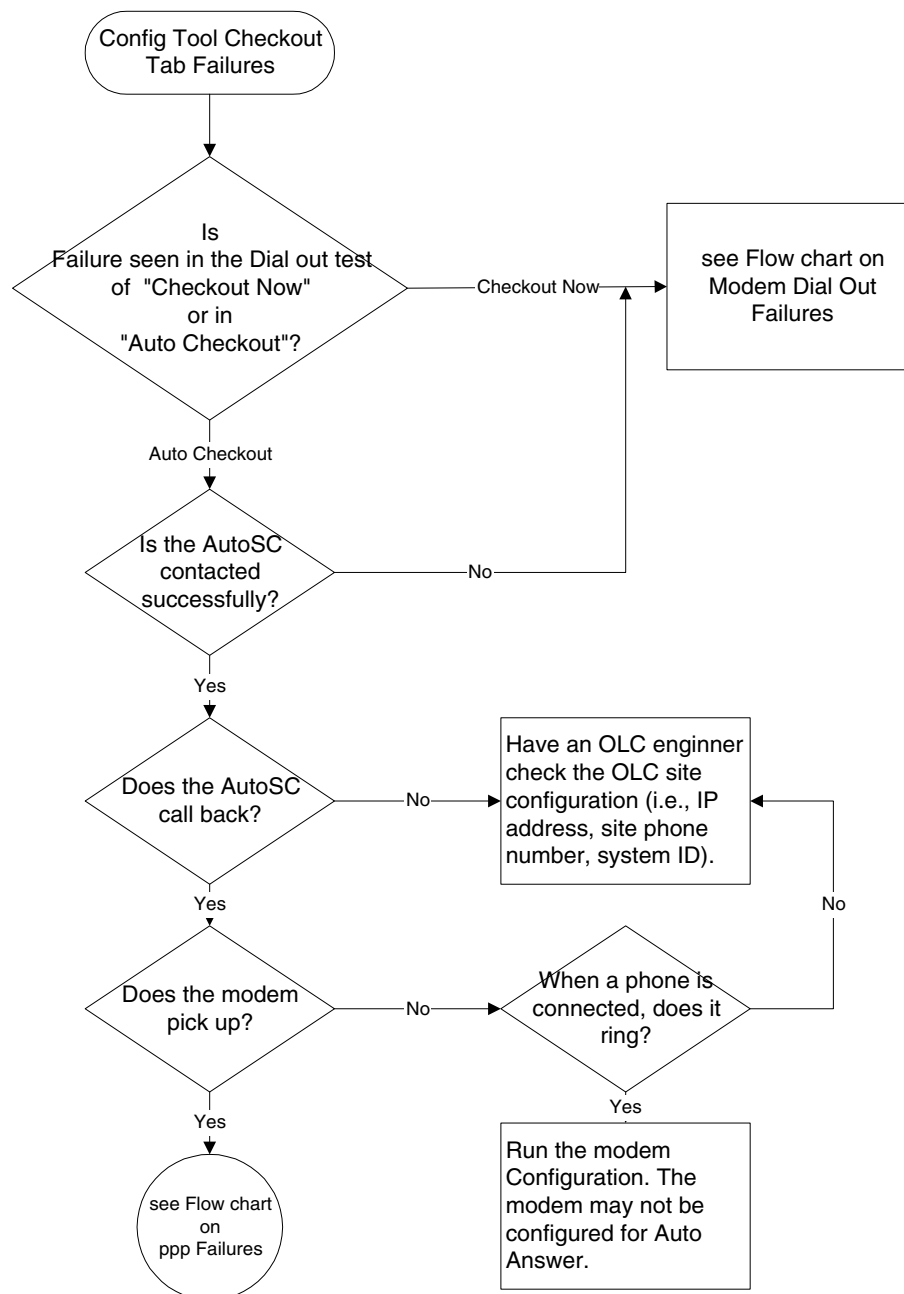


Figure 6-5 Config Tool Checkout Tab Failures - Flowchart

2.6 Communications between modem and CPU Failures

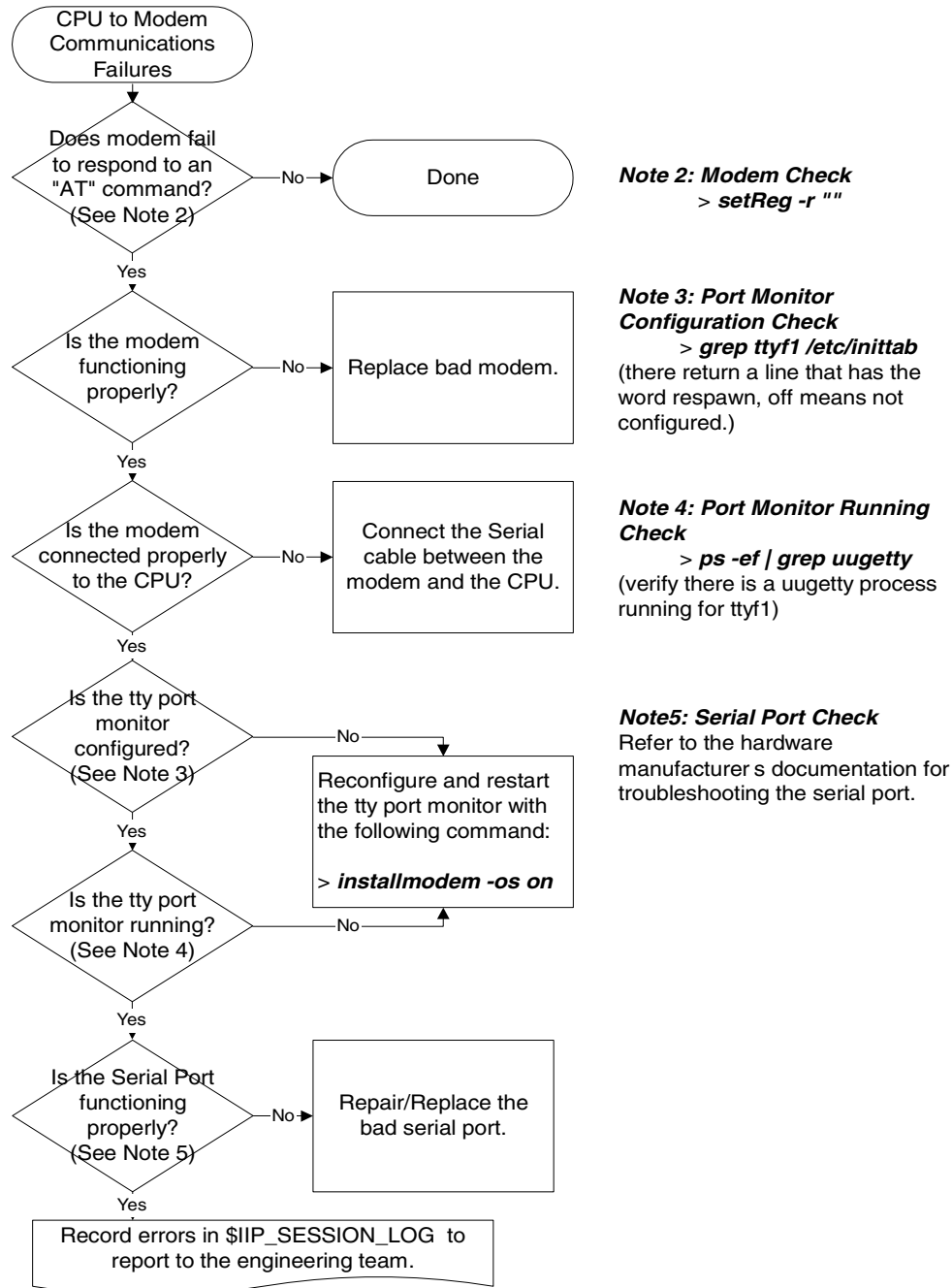


Figure 6-6 Communications between modem and CPU Failures - Flowchart

2.7 ppp Failures

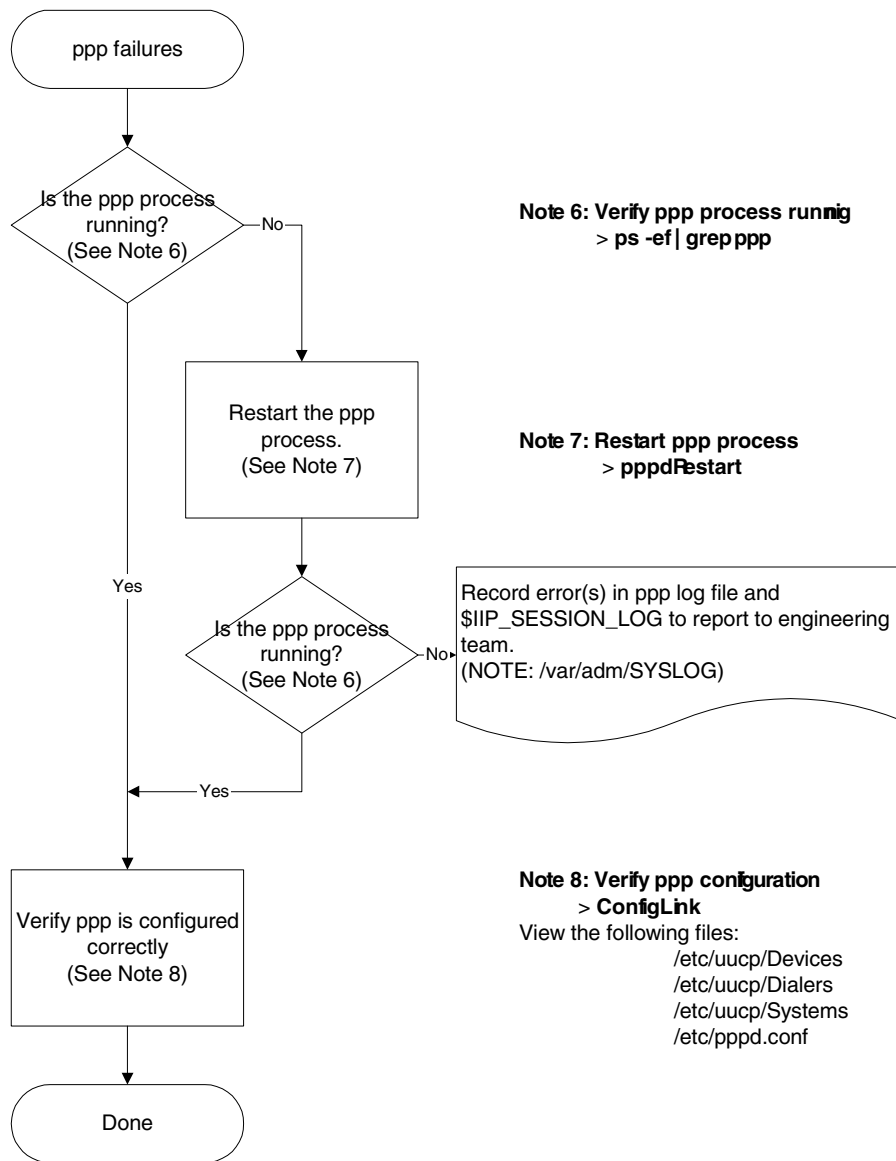
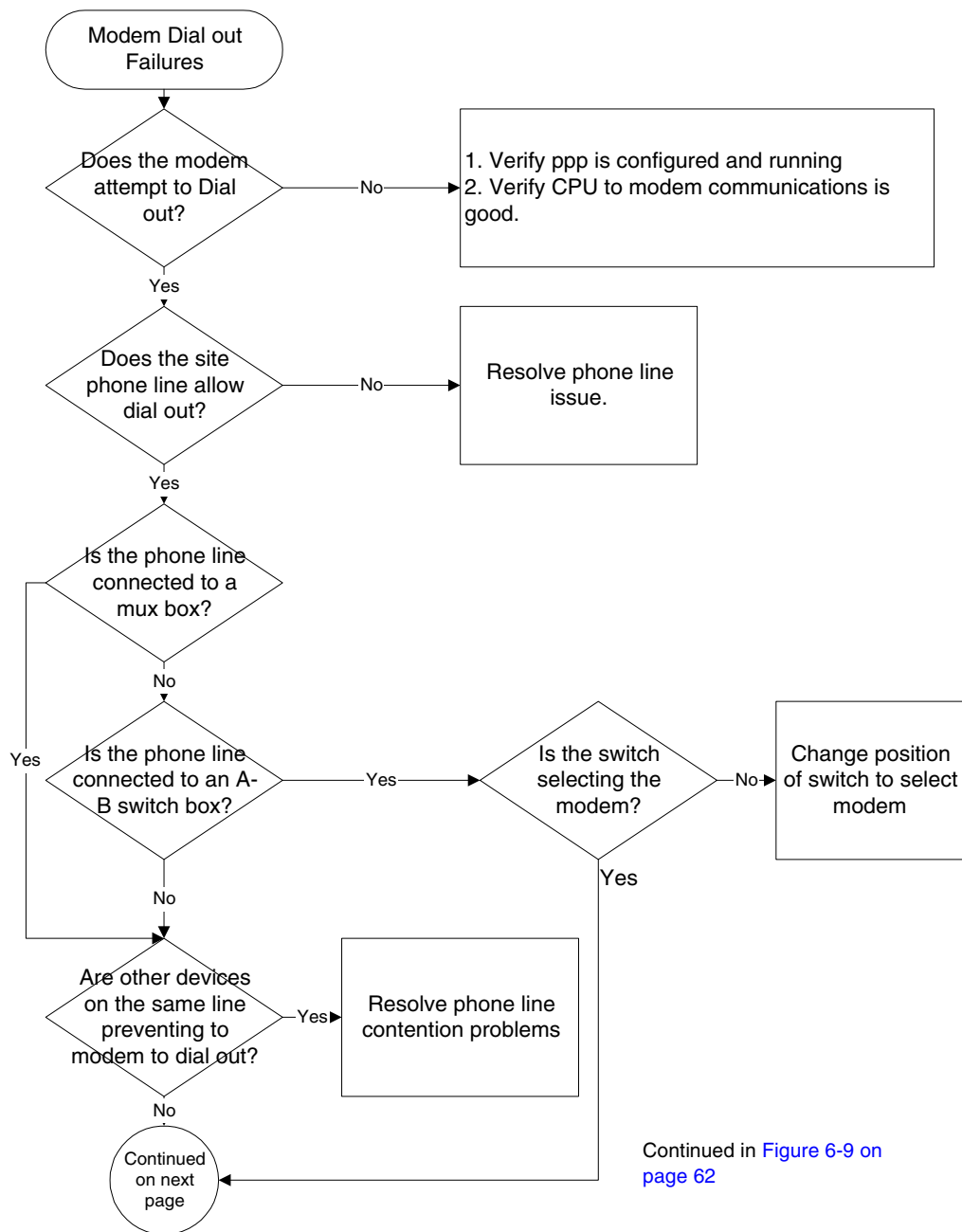


Figure 6-7 PPP failures - Flowchart

2.8 Modem Dial Out Failures



Continued in [Figure 6-9 on page 62](#)

Figure 6-8 Modem Dial-Out Failures - Flowchart

2.8 Modem Dial Out Failures (continued)

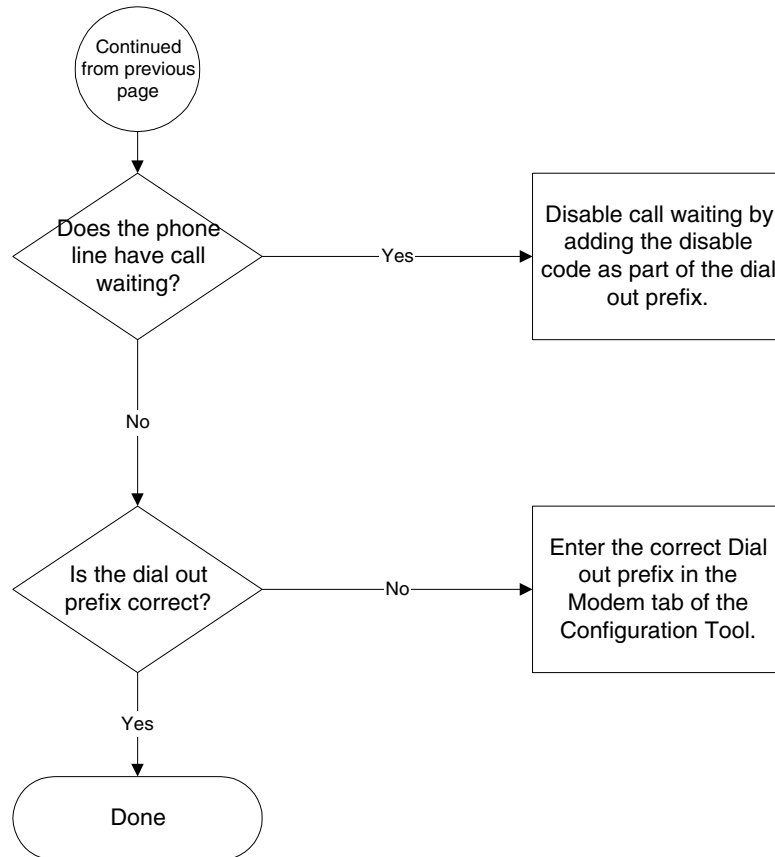


Figure 6-9 Modem Dial Out Failures - Flowchart (continued)

2.9 Web server problems

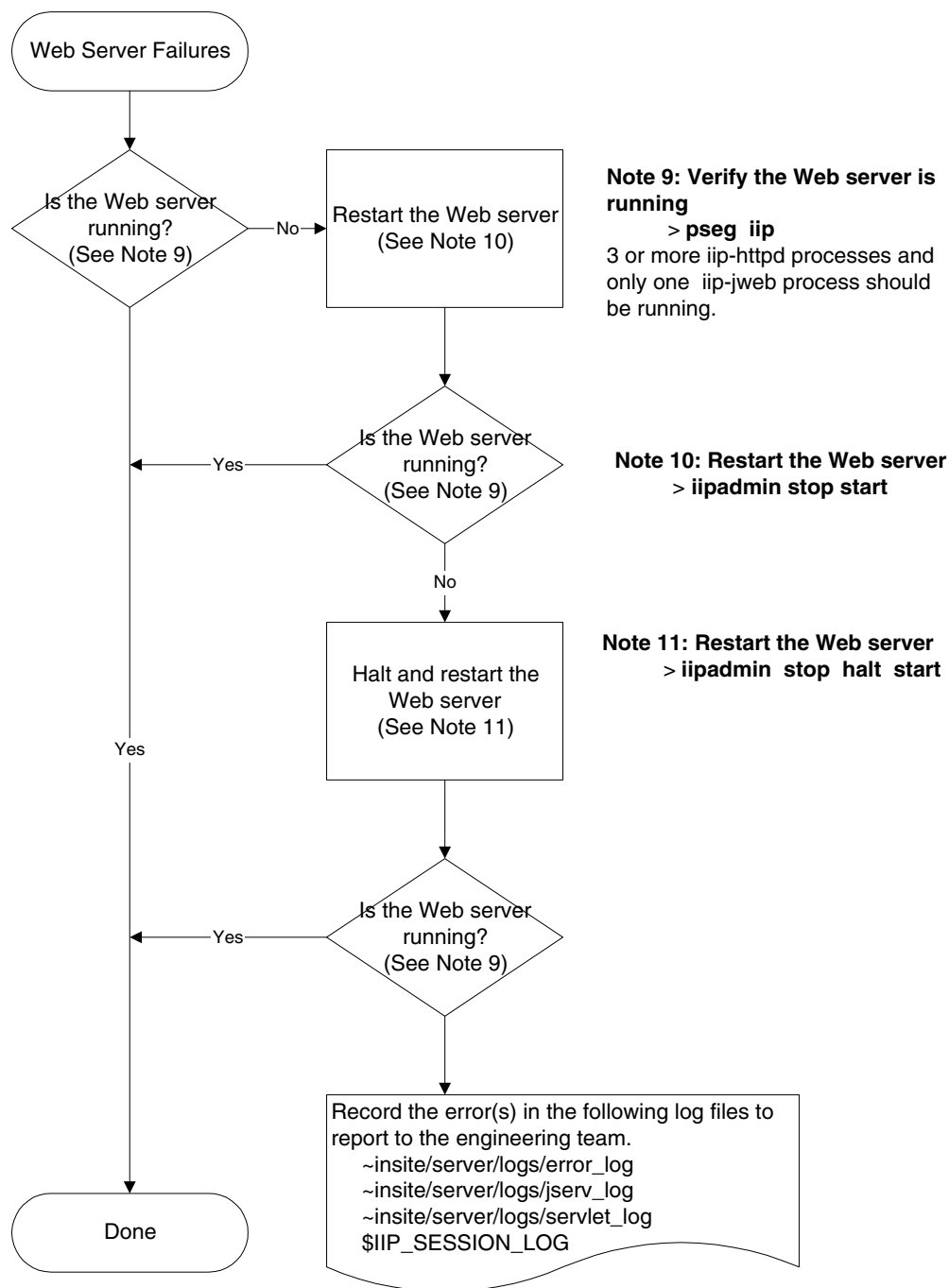


Figure 6-10 WEB Server Problems - Flowchart

2.10 Web applications won't run

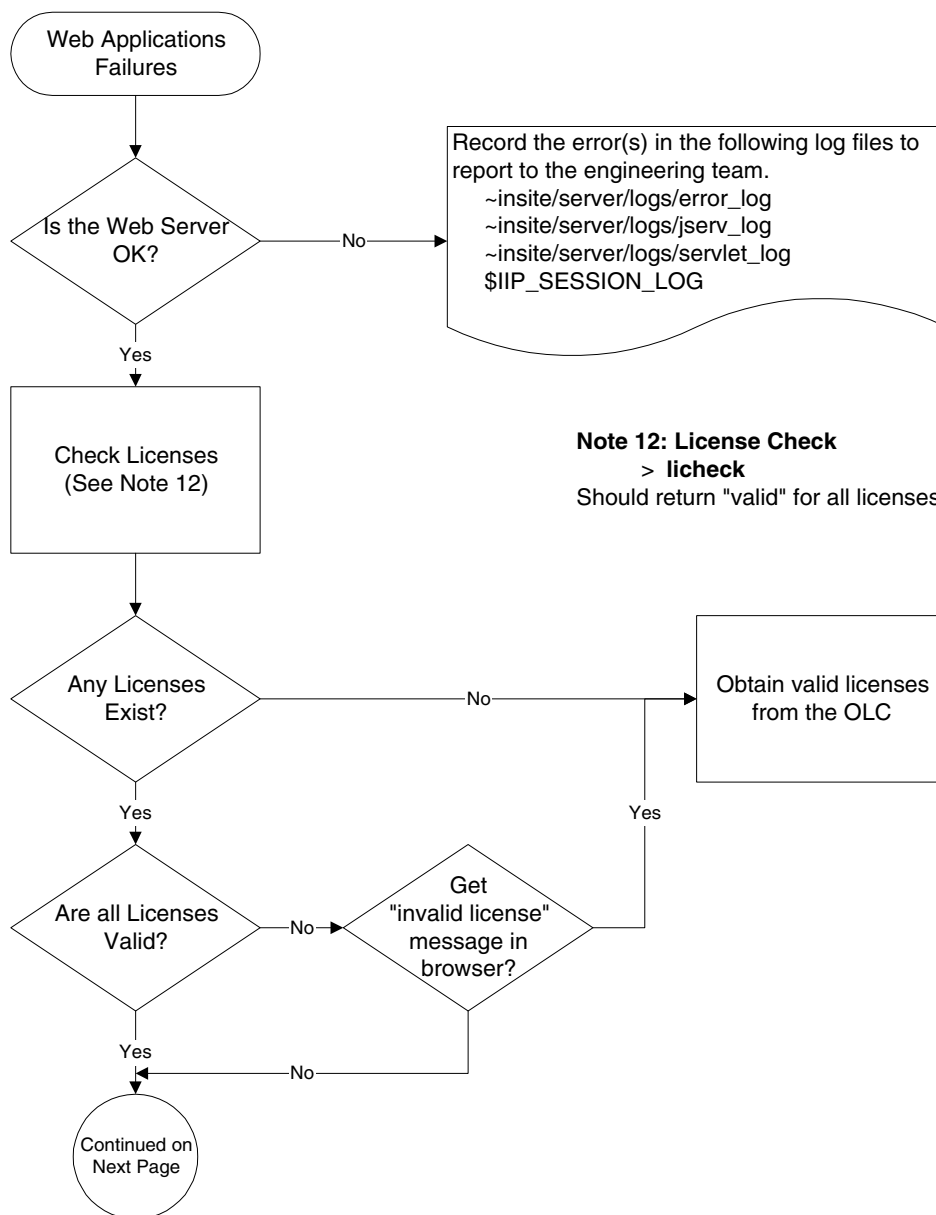


Figure 6-11 WEB Applications Won't Run - Flowchart

2.10 Web applications won't run (continued)

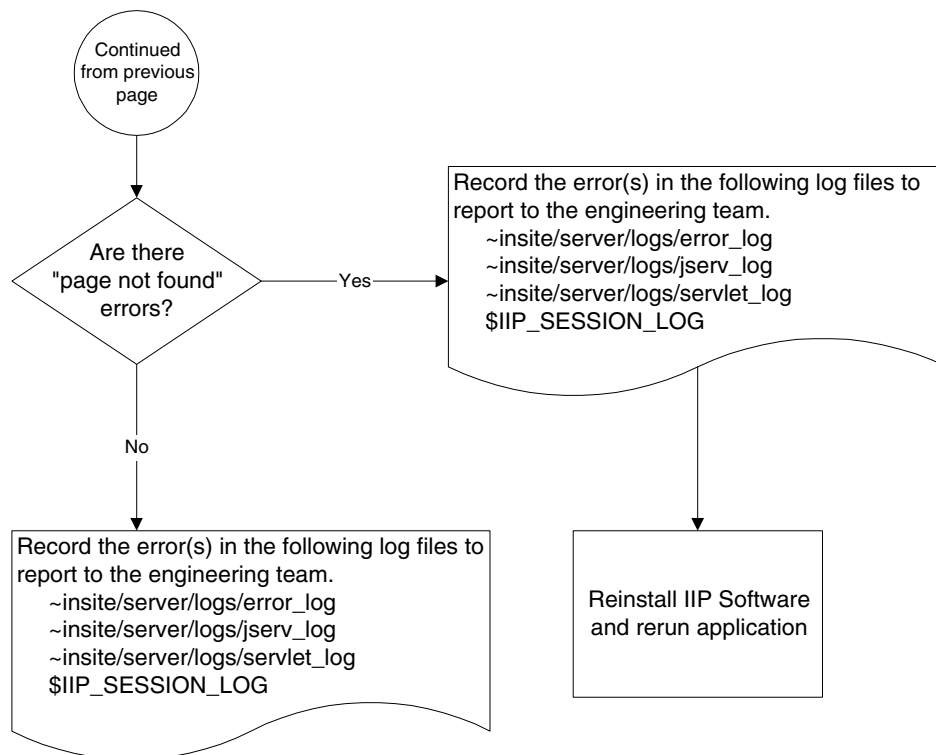


Figure 6-12 Web applications won't run (continued)

Chapter 7 InSite Features Descriptions

Section 1.0 Introduction

This chapter lists and describes the features of LightSpeed QXi InSite.

Section 2.0 InSite Dial-in Security Feature

The InSite Dial-In Security feature is for scanners that have optional InSite remote dial-in capabilities. This feature provides two levels of security:

- the CHAP encryption algorithm.
- a log-in/passwords for all accounts.

The On-Line Center (OLC) engineer will log in using the InSite Dial-In Security feature to call scanners in the field. If the scanner detects a failed log-in, it automatically generates an e-mail notice to the Automated Support Center (ASC).

Section 3.0 LASTFAIL

To list all failed log-ins, type `lastfail` ENTER

To list failed log-ins since a given date, type `lastfail`, followed by the date or time as in a Unix date command: `> lastfail Tue Dec 29 14:40:04` ENTER

Section 4.0 InSite Proactive Diagnostics Feature

The InSite Proactive Diagnostics feature is designed for scanners that have InSite remote dial-in capabilities. The Proactive Diagnostics feature automatically performs diagnostic testing of the scanner on a pre-set schedule. The scanner automatically dials the Automated Support Center to report the results of these tests. The results of the tests are forwarded to Support Engineers at the OnLine Center.

Each scanner has its own test schedule, during which testing can be performed. The field engineer should determine the normal working hours for the scanner, and when and whether proactive diagnostics should be run. A default schedule is supplied and should be fine for most systems. Most of the tests may run in the background while the operator is scanning. However, in the future, a test might require the use of the scanner. While this intrusive test is running, the plasma display shows a lockout screen. For emergency scanner use, the operator can interrupt intrusive testing by pressing the button on the lockout screen.

The field engineer selects the dial-out codes for calling the Automated Support Center (**ASC**). For example, find out whether or not you need to dial 9 for access to an outside phone line. The actual phone number for the **ASC** is downloaded to the scanner during InSite checkout.

Both the Field Engineer and the On-Line Center Engineer can use the **prodiag** command to perform these functions:

- Schedule tests.
- Execute tests.
- Turn tests off or on.
- Look at test results.
- Look at the log file.

4.1 Proactive Diagnostics Tests

All of these tests may run during normal system operation and should have no effect on the scanner. A default schedule is provided for these tests. The default schedule should be fine for a site, provided the OC is not shutdown or at the boot level. The schedule may be changed.

- **ChkLogs**: informs the ASC when selected error messages occur in specific log files.
- **flog**: informs the ASC when there have been five attempts to log in with the wrong password.
- **IIP_DialPrefix**: checks system log files for errors.
- **IIP_Housekeeping**: cleans up the IIP log files.
- **PassChg**: informs the ASC when an important password has changed on the system.
- **dialout_test**: dials the ASC to verify the out-going modem connection is working.
- **healthpg**: summarizes several selectable system performance parameters. Presents notes from the users.
- **pd_ASC_Notify**: calls the ASC when there is information to send.
- **pd_HouseKeeping**: deletes older files in the prodiags directory, trims the prodiags log.
- **slprune**: prunes the log report.

4.2 Recommended Schedule

- It is recommended that **pd_ASC_Notify** runs hourly.
- **pd_Housekeeping**, **flog**, **ChkLogs**, **ChkTube**, **IIP_DialPrefix**, **IIP_Housekeeping**, **slprune**, and **PassChg** should be scheduled to run daily.
- **dialout_test** should not be scheduled because it is intended to be run only as part of modem checkout.

4.3 Using the Prodiags Tool

This section discusses the prodiags tool. The field engineer can use this tool to modify the schedule or look at results. InSite can also use this tool. To start the tool, become superuser (**su** - and enter the root password), and then type: **prodiags**. A menu with these options appears:

1.) View Log

This displays the prodiags history log, a record of what tasks prodiags started and their success or failure. This log gets automatically pruned so that it will never grow beyond a preset size. When the log is pruned, the earlier entries are deleted.

2.) View Results

This allows you to see the result files from a specific task. First you would select the task that you are interested in, and then you select the proper result file. Some tasks do not create result files, so this option will say that no results are available. Note: to save disk space, prodiags

automatically deletes older result files. You can also delete result files by selecting the remove result option from the utilities menu.

3.) Execute a task

This is useful for testing prodiags. It lets you execute a prodiags task immediately. Normally, the task is scheduled to execute at a specific time in the future. When you select this option, you get a list of tasks to select from. When the task is selected, it will run to completion. Note that you won't see any output from the task while it is running because the tasks are designed to run without a person being there.

4.) Schedule management

This lets you view or modify the times when tasks are going to run.

- a.) **View Schedule** -- shows the current schedule.
- b.) **Modify schedule** -- lets you change the time for a task. For a background task, you can change the number of times a task runs (iterations), the time the task starts, and the day of the week that the task runs. You may also specify hourly or daily for the tasks to run. For intrusive tasks, you can change the time slot when the task runs. You set up the time slots on the Define TimeSlots menu. (A time slot is a period of time when the system is supposed to be idle.)
- c.) **Add task to schedule** -- lets you put in another instance of a task to a schedule. You can have a task scheduled to run several times, such as Mondays and Fridays. Enter the same information as when you Modify the tasks. Note: when adding an intrusive task, you should first define the time slot, and then add the task.
- d.) **Remove task from schedule** -- lets you remove a task from the schedule. (The task remains on the disk, so it can be added at a later date if desired.)
- e.) **Define Time slots** -- (Scanner Idle Time) -- lets you set up time slots when the scanner will be idle, modify those slots, remove those slots, and deactivate those slots. Modify Time Slots: Select this to modify or add a new time slot. Select a time slot to modify or "new" to define a new slot. You will be asked for the start and end time of the slot.
- f.) **Activate/Inactivate** -- An inactive time slot will execute. An inactive time slot will not run. You can in-activate a time slot if you want to keep the tasks in that slot defined, but you don't want the slot to run. For example, if the schedule at the site temporarily changes.
- g.) **Delete a Time Slot** -- This removes the time slot from the schedule.

5.) Utilities

These are additional tools for expert users to manually prune or delete files (if you need disk space immediately), install a new task, or remove an existing task, or view configuration files.

- a.) **View Task Info** -- shows configuration information for the task.
- b.) **View Config File** -- shows configuration information for the prodiags tool.
- c.) **Install a Task** -- Install a task that was loaded by a patch tape or a software download.
- d.) **Setup a Task** -- use to add or delete messages for which to check.
- e.) **Remove a Task** -- this deletes the task from the disk.
- f.) **Prune Log** -- manually prune the log.
- g.) **Prune All Results** -- manually prune results for all tasks.
- h.) **Prune Results for Task** -- manually prune results for a specific task.
- i.) **Delete Results** -- remove any result files from the disk.

Section 5.0 System Health Page Feature

5.1 Overview

The health page provides quick access to logs and statistics used during planned maintenance and troubleshooting. The health page contains a collection of logs and statistics, gathered over a specified time period (typically 30 days.) The system automatically generates the health page and mails it to up to ten users, per a configured schedule. Schedule a health page generation every 30 days, or customize the schedule to the system PM dates.

Configure the health page during IIP Configuration. Once you install the health page, the system automatically mails the reports to the designated addresses, and requires no additional user interaction. The Health Page also has a customer SERVICE NOTE PAD to allow them to create messages for the FE. This feature can be used to give the FE a reminder of items to check on the scanner. The System State Characterization Files selection saves the health page configuration.

The system generates the health page report three days prior to the customized dates, in order to mail out the report in time for the PM. If the scanner happens to be down during the scheduled reporting time, the system mails the report as soon as it is back up. If you chose to customize the schedule, and all the dates you entered have passed, the system automatically generates the report every 30 days, until you enter new dates into the schedule.

In addition, you may generate the health page from the PM menu in the service desktop, from InSite, or by sending an e-mail request to the Automated Support Center, asking them to send a copy to you. The ASC also retains a copy of the full report for future trending purposes.

Use the healthpage tool to set the content of the health page, the schedule for sending the report, designate the report recipients, and generate or view a report. The healthpage tool is available on the proprietary service desktop. Remote users can also use the healthpage command.

The health page contains the following information for a specific time span (typically 30 days.)

- **System Hardware and Software Configuration:** Includes the network addresses, software revisions, system ID, hospital name.
- **Number of InSite Logins:** The number of InSite logins to the system will be shown.
- **Number of computer reboots:** The number of times someone restarted the OC software and the date of the last five system reboots.
- **Number of scan hardware resets:** The number of times scan hardware required a reset, and the number of times the reset failed.
- **Generator Errors:** The first line of any Primary/Most Severe error message, for the generator subsystem.
- **Primary/Most Severe Error Summary:** The first line of the most frequent primary/most severe error messages from the system log. The software sorts the messages by frequency, and lists the first 25 (you can configure this number) in the report.
- **Notes for Service:** Notes that InSite added to the session log or customers added to the message log, to inform you about problems that arose since the last report.
- **Selected Errors from Other Logs:** The results of a search of the /var/adm/messages logs on the OC and SBC for failures and panics. Only the last ten are displayed.
- **Applications Core Files:** A list of core files on both the OC and the SBC.
- **Tube Spits:** The tube spits report from viewStats.
- **RCom/SCom retry rate:** The statistics from viewStats.
- **Axial Motor Current:** The statistics from viewStats.
- **Rotor Uptime:** The rotor uptime statistics from viewStats.
- **Temperatures:** Gantry, OBC, and Detector temperature statistics.
- **Converter Board Temperatures:** Min and Max DAS Converter Board temperatures.

- **Tube Usage:** The tube usage report from viewStats, for the time period covered by the health page report.
- **Scan Usage:** Lists the number of scans and number of slices with a given kV/mA, Scan Time, Scan Mode, Focal Spot, and Usage Mode.

5.2 Accessing the Health Page Report

Access the health page report one of these ways:

- **Automatically**

Program the scanner to automatically create the health page report, per the schedule entered during installation, or at a later time. The system mails the health page to the configured addresses. You can select and view the health page report in the proprietary service desktop under the PM menu.

The automatically generated report covers the time interval since the last report.

- **Remotely - via InSite**

If you want immediate access to the health page report, direct the scanner to generate it for you. Open a shell window, and type/enter `healthpage`, to create the health page report while you wait. (It can take up to 6 minutes for a full report.) Upon completion, the system displays the report on the screen.

To override the customized report content, type/enter `healthpage -f` to display the full report.

To display the previously generated health page, type/enter `healthpage -s`.

The manually generated report covers the preceding 30 days (unless you configured more or fewer days.)

- **On Site - via the Service Desktop**

If you want immediate access to the health page report, direct the scanner to generate it for you.

On the proprietary SERVICE DESKTOP, under PM, select SYSTEM HEALTH PAGE. This will bring up the GUI version of the Health Page tool.

- **By requesting it in e-mail**

You may request the current healthpage report by sending an e-mail to the Automated Support Center (ASC). The ASC will SWEEP the system for the report, and e-mail it to you in under 30 minutes. This report covers the preceding 30 days (unless you configured a different number of days) and contains the customized content.

Anyone with a GEMS email address can request a report. You do not have to have your address in the email list on the scanner.

From your mail tool, create an e-mail message with the following content:

- **From:** (your e-mail tool fills this in for you)
- **To:** healthpg@iscmed.med.ge.com
- **Subject:** Health Page (Include the following information in the mail message:
- **USN =** XXXXXXXXXXXX
- **SYSTEM ID =** XXXXXXXXXXXX (USN & SYSTEM ID should match the system of interest.

5.3 Health Page Tool

The health page tool is accessible under the PM list on the proprietary SERVICE DESKTOP. The health page GUI allows the user to generate and configure the health page report after installation.

The health page tool has three menu items:

- **Exit:** Exit the Health Page Tool.
- **Report:** Generate and View the report.
- **Configure:** Configure the Health Page content and schedule.

The **Exit** selection exits the tool.

The **Report** selection displays the following pull-down menu:

- **Custom Report:** Generate a Customized Report according to the Content settings.
- **Full Report:** Generate a complete Health Page report.
- **Last Report:** View the Last Report generated.

If **Custom Report** or **Full Report** is selected, a message appears in the display window saying that the health page is being generated.

When the report is available, the message is removed and the health page is displayed, (Refer to [Figure 7-1](#)). The path name to the file is displayed in the bottom center portion of the screen. Enter search text and button to Search. The Search tool is initially not displayed until the report is displayed.



Figure 7-1 Health Page Report

The **Configure** selection displays the following pull-down menu:

- **Disable/Enable Health Page:** Disables/Enables Health Page Execution
- **Edit Schedule:** Define the runtime schedule for Health Page.
- **Edit Email Addresses:** Define users who should receive the Health Page report.
- **Edit Report Content:** Customize the content of the Health Page report.

5.4 Remote Users: The healthpage Tool

InSite, remote, and on-site users can use a command line version of the healthpage tool. This has the same capabilities as the GUI version. Open a shell window on site, or access the system from InSite, and use the following commands.

- Note:** Use this tool to manually generate or configure the report. You do not have to use the tool to generate an automatic report or to have one e-mailed to you.
- **healthpage -h:** Displays the list of health page commands.
 - **healthpage:** Creates a new health page from the previous 30 days (or whatever you configured) and displays it on the screen. The report only contains the configured sections.
 - **healthpage -f:** Creates a new health page from the previous 30 days (or whatever you configured) and displays it on the screen. The report contains ALL sections regardless of the current configuration.
 - **healthpage -v:** Displays the last health page report on the screen. Use this function when you just created the health page report and want to view it again.
 - **healthpage -off:** Disables the automatic generation of the health page report.
 - **healthpage -on:** Enables the automatic generation of the health page report.
 - **healthpage -c:** Customizes the health page report. (See section 5.5.)

5.5 Configure the Health Page Report; healthpage -c

You may customize the content of the health page report by turning some logs on or off, specifying the e-mail addresses to use, and setting the health page schedule.

During IIP Configuration, the system prompts for information to configure the health page report. See “Custom Health Page Instructions Tab” on page 38. The default values and the answers gathered during installation should be sufficient to run health page.

After installation, you may modify the health page configuration by typing/entering the command, **healthpage -c**, to display the following menu:

```
Configuring health page...
Health Page Configuration Main Menu
(P)M Schedule
(E)mail list
(R)eport configuration
(Q)uit
```

- Type **P** or **p** ENTER to enter or modify dates to generate the health page report.
- Type **E** or **e** ENTER to enter or modify the list of people who will be mailed a health page report.
- Type **R** or **r** ENTER to modify the report configuration.
- Type **Q** or **q** ENTER to quit.

5.6 Change the PM Schedule

Type/enter **P** from the Health Page Configuration Main Menu to display the following **Configured PM Schedule**. The system generates a health page report 3 days prior to a planned maintenance date. You do *not* have to enter PM dates; you may accept the default schedule of every 30 days.

Configured PM Schedule

(E)dit/Add

(D)eleate

(Q)uit

- Type **E** or **e** ENTER to enter or modify dates to generate the health page report.
- Type **D** or **d** ENTER to delete dates.
- Type **Q** or **q** ENTER to quit.

If you type/enter **E**, the system displays all of the currently configured dates, and then prompts you with each of the previously configured dates. For example:

PM date [05/16/99] :

Press ENTER to keep this date or type in a different date.

After the last available date, the prompt changes to:

PM date [new date] :

Type in the new date or press ENTER to exit.

You may enter as many dates as you want.

If you type/enter **D**, the system displays each of the previously entered dates, and prompts you to keep or delete it.

5.7 Change the E-mail Addresses

Display the Health Page Configuration Main Menu, and type/enter **E** to change the e-mail address(es). You may send the health page report to up to ten addresses. Enter the addresses during IIP Configuration, and use this command to change them.

Make sure you type in the full e-mail address. Otherwise, the system cannot deliver the mail.

The system displays the list of current e-mail addresses, and the following selections:

(E)dit/Add

(D)eleate

(Q)uit

- Type **E** or **e** ENTER to enter or modify the e-mail addresses.
- Type **D** or **d** ENTER to delete addresses.
- Type **Q** or **q** ENTER to quit.

If you type/enter **E**, the system displays each of the previously entered addresses. For example:

Example : name@med.ge.com

Enter an address or press return to keep the current value.

address[john.smith@.med.ge.com] =

Press ENTER to keep this address, or type/enter a different address.

After the last available address, the prompt changes to: address [new address] :

Type in the new address, or press **ENTER** to exit. You may enter up to ten addresses.

If you type/enter **D**, the system displays each of the previously entered addresses, and prompts you to keep or delete it.

5.8 Change the Report Content

Display the Health Page Configuration Main Menu, and type/enter **R** to change the health page content. Use this menu to change the frequency and content of the health page report. Use this function to disable the sections of the report that you do not want to see.

Note: If you configured the report during IIP Configuration, you do not have to modify it.

The system uses your selections during automatic report generation.

Type/enter **R** to display the following menu:

Configuring health page...

1. Number of Pri/Most Severe errors to display? [25]
 2. Number of days to report? [30]
 3. Number of days prior to send report? [3]
 4. Include System HW and SW Configuration? [Y]
 5. Include Number of InSite Logins? [Y]
 6. Include Number of Computer Reboots? [Y]
 7. Include Scan Hardware Resets? [Y]
 8. Include Generator errors? [Y]
 9. Include Instances of Pri/Most Severe Errors? [Y]
 10. Include Notes for Service? [Y]
 11. Include Selected Errors from other Logs? [Y]
 12. Include Applications core files? [Y]
 13. Include Tube Spits? [Y]
 14. Include RCom/SCom retry rate? [Y]
 15. Include Axial Motor Current? [Y]
 16. Include Rotor Uptime? [Y]
 16. Include Temperatures? [Y]
 17. Include SelectedErrors from other Logs? [Y]
 18. Include Tube Usage? [Y]
 19. Include Scan Usage? [Y]
- 'q' to quit Enter Option:

Select the options you want to change. Type/enter **q** to exit this menu.

Section 6.0 Service Notepad Tools

The customer, or InSite engineer, or another FE can use this function to leave notes about items they want the primary FE to address during a PM visit. Display the healthpage report or the message log to view the customer notes, and display the healthpage report or the session log to view the InSite and the FE notes. There are a number of different ways for the Customer, the InSite Engineer, and the FE to leave notes in the message log, the session log, and in the healthpage "Notes for Service". Each of these methods of communication is discussed in this section.

6.1 Customer Messages

The Customer can easily leave messages for the FE or for InSite to read, by using the MEMO button on the messages screen. This function will leave a message in the GE System Log, labelled as added by the OPERATOR. These messages will also appear in the healthpage report in the Notes for Service, Customer Notes section of the report. The notepad tool will prompt the user to type in a string to be placed in the message log.

Enter a note to put in the message log and select save.

In addition, anyone can enter an "Operator Comment" by using one of the following commands (open a UNIX Shell and type one of these commands):

- **SvcNotepad -o**: The user will be prompted to input a message to log, and press ENTER to update and exit.
- **Notepad**: This will bring up a "GUI" that will direct the user to enter a note and click on the SAVE button to save the message to the message log.

6.2 InSite Messages

The InSite Engineer can leave messages in the session log for the FE to read. The command listed below will leave a message in the session log, labelled as an InSite Comment. These messages will also appear in the healthpage report in the Notes for Service, InSite Notes section of the report. To input an "InSite Comment", type the following command (open a UNIX Shell and type):

SvcNotepad -i:

The user will be prompted to input a message to log, and press ENTER to update and exit.

6.3 FE Messages

The FE can leave messages in the session log for an InSite Engineer, or another FE to read. This function can be used to pass messages to another FE, when more than one person performs service on the same machine, or to pass messages to an InSite Engineer.

The FE can easily leave messages for another FE or for InSite to read, by using the SERVICE NOTEPAD selection on the SERVICE DESKTOP. This function will leave a message in the session log, labelled as an FE Comment, and if one takes the time to enter their name in the space provided on the Service Notepad GUI, the message will be labelled as From <your name>. These messages will also appear in the healthpage report in the Notes for Service, FE Notes section of the report.

In addition, to input an "FE Comment" from within a UNIX Shell, type: **Notepad -SL**:

This will bring up a "GUI" that will direct the user to enter a note, enter their name, and click on the SAVE button to save the message to the session log.

Section 7.0 Remote Boot Feature

The remote boot feature gives the InSite engineer the ability to connect to a scanner that is having trouble bringing up UNIX, to boot it, or to run standalone diagnostics. The remote boot feature redirects all input and output from the console to the modem port. While remote boot is active, there is no input or output from/to the console keyboard/monitors. The modem is directly connected to the EAW during remote boot. This is not a PPP connection. The customer or on-site engineer must enable remote boot mode. This is done by typing in some commands at the console. Then the EAW can connect to the scanner. When the InSite engineer is finished, it will be necessary for someone at the site to re-boot the computer.

7.1 Pre-Check for Remote Boot

Once remote boot is enabled, all input and output goes to the modem port. If the modem is unable to connect to the port, a service call will be required to set the input and output back to the console. Although this is a rare occurrence, we suggest that the modem connection is tested out before actually starting up a remote boot session.

See Troubleshooting for a procedure on how to recover if this occurs.

Note: **VERIFY THE MODEM CONNECTION FIRST! THIS IS VERY IMPORTANT TO DO.**

7.1.1 Verify modem connection

- 1.) If you have a PPP connection to the site, terminate that connection and wait a few minutes for it to time out, and for the modem to hang up the line.
- 2.) At the EAW, select BOOT PORT DIAL button on the connect tool menu. This will try to use the Portmaster Boot Port modem to call the modem at the scanner. The modem at the scanner will not answer the phone, but you should verify that the phone at the site is being dialed. If so, you may proceed with remote boot.

7.2 On-Site Setup for Remote Boot

You will need the customer or an FE at the scanner to perform these steps. The InSite engineer should explain to the customer what to select or type at this point. The step to take at this point depends on the state of the system:

- 1.) If the scanner is at applications level, instruct the user to shutdown applications.
 - a.) If the scanner is at UNIX level, then shutdown UNIX, (switch user to root) :
 - * `su -` ENTER (and enter the root password)
 - * `/etc/halt` ENTER
 - b.) Select the RESTART button.
 - c.) Proceed to step 3.
- 2.) If the scanner will not boot UNIX, then instruct the customer to either cycle the console power or press the reset button on the computer.
- 3.) Select the STOP FOR MAINTENANCE button.
- 4.) Select ENTER COMMAND MONITOR from the System Maintenance Menu.
- 5.) At the next prompt, type in the following commands:
 - `setenv console d` ENTER
 - `exit` ENTER
- 6.) Select START SYSTEM from the System Maintenance Menu.

The computer will now send all of its output to the modem. No input or output will be seen on the console monitors at this point until remote boot is terminated.

7.3 Remote Setup for Remote Boot

The remaining steps are done from the EAW.

- 1.) At the EAW, select BOOT PORT DIAL button on the connect tool menu. This will use the Portmaster Boot Port modem to call the modem at the scanner.
- 2.) Once the modems connect, press ENTER a few times.

You will see the following menu:

```
System Maintenance Menu
1) Start System
2) Install System Software
3) Run Diagnostics
4) Recover System
5) Enter Command Monitor
```

7.4 To Run Diagnostics

- 1.) Select item (3) from the System Maintenance Menu.
- 2.) Press the <ESC> button to return to the System Maintenance Menu.
- 3.) Refer to the System Service Manual (2211222-100) for details of how to run and how to interpret the IDE Diagnostics tests.

7.5 To Boot UNIX

- 1.) Select item (5) from the System Maintenance Menu.
- 2.) Type: `sash` ENTER
- 3.) Type: `unix` ENTER

UNIX should come up and the startup messages should be displayed at the EAW. The login prompt will appear. With UNIX running, you should be able to perform any UNIX commands.

7.6 Ending a Remote Boot Session

Note: **VERY IMPORTANT -- DO NOT SKIP THIS STEP!**

When you are done with remote boot:

- 1.) Hang up the EAW connection.
- 2.) Have someone at the site re-boot the scanner.

Remote boot should be ended now. You can now connect via ppp. The user should see output on the console.

7.7 Troubleshooting - Modem failure during Remote Boot

If the modem fails or the serial ports or cable fail after the computer has been set up for remote boot, after Section 7.2 step 5.) above, the computer can become stuck in remote boot mode. The

symptom is that the keyboard does not respond, and there is no display on the console monitors and the EAW cannot connect to the system. If the EAW fails to connect, try connecting to the scanner using a laptop computer with a modem, and proceed to step 5.).

If you still fail to connect, follow the procedure below starting with step 1.).

The following procedure can be performed at the scanner to fix this problem. You will need an additional cable to hookup a PC/Laptop to the SGI system. The cable required is a NULLMODEM cable with a male 9 pin, D-shell connector (called a DB-9 connector) on one end and a male DB-9 connector at the other end.

- 1.) Connect one end of the special NULLMODEM cable (male DB-9 connector) to the PC COM1 PORT.
- 2.) Disconnect the cable from the back of the OCTANE Computer (serial port #1 - where the modem cable is presently connected) and connect the NULLMODEM cable's other end DB-9 connector.
- 3.) Configure the laptop to be a terminal emulator for the modem port. For example, you can use the Windows 95 windows terminal program (terminal.exe) with the following settings:
 - Terminal Emulation - set to DEC VT-100 ANSI
 - Communications
 - * Baud Rate : 9600
 - * Data Bits : 8
 - * StopBits : 1
 - * Parity : None
 - * Flow Control : xon/xoff
 - * Connector : COM1
- 4.) Cycle power or push the system reset button on the SGI.
- 5.) Press **ENTER** a few times to see the output from the SGI on the PC terminal window.
- 6.) When the system maintenance menu comes up, select **ENTER COMMAND MONITOR**.
- 7.) Type: `setenv console g` **ENTER**.
- 8.) Cycle power or push the system reset button. After the system comes up, you should see all output at the console, and the keyboard should be operational.
- 9.) Disconnect the laptop and reconnect the modem.

7.8 Troubleshooting - Modem failure AFTER remote boot completes.

If the remote boot feature was run successfully, but you are having difficulty bringing up the PPP connection try the following:

- 1.) Cycle power on the modem.
- 2.) Perform the IIP Configuration procedures contained in [Chapter 3](#), specifically the modem configuration procedures section, to reprogram the modem for normal operation.

Section 8.0 InSite Logged In Feature

When InSite connects, a message is posted to the current messages screen.

If the system has the optional InSite Interactive license installed, the InSite icon on the desktop will also change as explained in [Chapter 1](#).

Section 9.0 Modem Back Door, Non-IP Connection

9.1 Overview

Modem security has been designed in the QX/i system such that only the PPP/Chap connection will be available. In anticipation that there could be an occasional problem with the PPP connection, a Modem Back Door has been included. The Modem Back Door was designed to be controlled by the customer. If the PPP connection fails, the OLC could request the customer to open the Modem Back Door. The OLC would walk the customer through the steps; the OLC would then dial in to the site (using a non-PPP connection) and enter a username and password. When the Modem Back Door is opened, the OLC will have ten minutes to connect. Once connected, the OLC or customer can close the back door and the current connection will remain intact. Note that both the opening and closing of the back door will cause an ATTENTION pop-up window to be displayed to the operator informing them of the state of the Modem Back Door. It is important to have the operator's approval before changing the state of the back door.

9.2 MBD Procedure

The OLC should walk the customer through the following steps:

- 1.) Select the SERVICE DESKTOP
- 2.) Open a UNIX Shell Window
- 3.) Type: `mbd open` ENTER

Note: Note that an ATTENTION window will be displayed to the operator. When the customer completes these steps, the OLC has ten minutes to connect to the site.

The OLC should dial into the site (using a non-PPP connection) and enter a username and password. Once in, the OLC or customer can close the back door using the following step:

- 4.) Type: `mbd close` ENTER

Note that another ATTENTION window will be displayed to the operator.

9.3 MBD Tool Specifics

The tool is located in ctuser's, insite's, and service's path (but not root's); it is an SUID script:

```
{ctuser@rhapl}[1] which mbd ENTER
/usr/g/ctuser/bin/mbd
{ctuser@rhapl}[2] ls -l /usr/g/bin/mbd ENTER
-r-sr-xr-x 1 root sys 7104 May 6 09:08 /usr/g/bin/mbd*
```

9.3.1 MBD Tool Syntax and Example

Entering just the tool name will display both the Syntax and an Example:

```
{ctuser@rhapl}[4] mbd ENTER
Syntax: mbd [-h|help|info|status|open|close]
Example: mbd info
```

9.3.2 MBD Tool Info

Entering the example will display information about the tool:

```
{ctuser@rhap1} [5] mbd info ENTER
```

=====

=== Modem Back Door Script ===

This GEMS product was designed with strict modem security. Normally modem access is disabled unless InSite is purchased. If InSite is installed, ONLY the InSite user/password is allowed access; in addition, a second level of security challenge must also be passed. This double level of security keeps the system fairly secure.

This script was designed as a precautionary measure. If InSite should ever have a problem connecting to this product, the customer may run this script to open the modem to a single level of security for a few minutes; i.e.: a caller must connect within a few minutes of this script being run and then the caller must pass the InSite user/password challenge -- the second level challenge would not be issued.

This tool puts control of the modem back door in the customer's hands.

Disclaimer: GE Medical Systems has provided the best measures it can for modem security; it has no control over Third Party or client system administrator's modifications. Ultimately, security always lies in the hands of the customer.

=====

If you have not already done so, please read the 'help' description by typing: `mbd help`.

```
### The current Modem Back Door status is: 'CLOSED'.
```

9.3.3 MBD Help

Entering the help command will display the help description:

```
{ctuser@rhap1} [6] mbd help ENTER
```

```
=== Modem Back Door Script ===
```

Syntax: mbd [-h|help|info|status|open|close]

Example: mbd info

```
-h ..... Display this help menu.
```

help Also, displays this help menu.

```
info ..... Describes this tool.
```

```
status ... Displays the current Modem Back Door
           status (ie: opened/closed).
```

```
open ..... Opens the Modem Back Door.
```

```
close ..... Closes the Modem Back Door.
```

Note: The 'open' command will open the back door and spawn a job to close the back door after 600 seconds.

If you have not already done so, please read the 'info' description by typing: `mbd info`

```
### The current Modem Back Door status is: 'CLOSED'
```

The back door will need to be manually closed (using the `mbd close` command if the system loses power or is rebooted prior to execution of the automated close.

9.3.4 MBD Tool Status

Entering the status command displays the current state of the Modem Back Door:

```
{ctuser@rhapl}[7] mbd status ENTER
```

```
### The current Modem Back Door status is: 'CLOSED'
```

9.3.5 MBD Tool Open

Entering the open command will open the Modem Back Door for 10 minutes and display an ATTENTION pop-up window to the operator:

```
{ctuser@rhapl}[8] mbd open ENTER
```

```
Opening the Modem Back Door ...
```

```
(The back door will remain open 600 seconds.)
```

```
### The current Modem Back Door status is: 'OPENED'
```

Note: The back door will remain OPEN if the system is reset or loses power prior to the automated close; if this occurs, the customer should manually close the back door after the reboot.

9.3.6 MBD Tool Close

Entering the close command will close the Modem Back Door and display an ATTENTION pop-up window to the operator:

```
{ctuser@rhapl}[9] mbd close ENTER
```

```
Closing the Modem Back Door ...
```

```
### The current Modem Back Door status is: 'CLOSED'
```

Chapter 8

GLOSSARY

LISTING OF TERMS

AutoSC - Automated Support Center. The automated support center accepts incoming requests from ProDiags or call out to sites to gather information.

CPU - Central Processing Unit.

daemon - Disk And Execution MONitor. A UNIX process that continuously runs to monitor or process requests or commands of a specific type.

GUI - Graphical User Interface.

HTTP - Hypertext Transfer Protocol. Advanced protocol for the exchange of hypertext documents across the Internet, with destinations of all internal and external hyperlinks included.

II - InSite Interactive. This is the big program. Parts include the InSite Interactive Platform, customer applications provided by modalities, and applications and reports provided by the OnLine Center.

IIP - InSite Interactive Platform. This includes InSite connectivity, Proactive Diagnostics, Web Server, Web Browser, and Customer Applications.

IP - Internet Protocol. A layer three protocol used in a set of protocols to support a network. IP provides a connectionless datagram delivery service for transport layer protocols such as TCP.

OLC - OnLine Center.

PPP - Point-to-Point Protocol. A TCP/IP protocol used over a modem connection.

ProDiags - Proactive Diagnostics. A GEMS proprietary tool that runs small diagnostic programs to proactively identify a potential site problem. ProDiags notifies the OLC when a diagnostic identifies a "known" error.

SGI - Silicon Graphics Inc.

TCP - Transmission Control Protocol. TCP is the virtual circuit protocol of the Internet protocol family. It is a byte-stream protocol layered above the Internet Protocol(IP).

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Appendix A

InSite Interactive Command Summary

Command line parameters that may be added to the “iipadmin config” to customize the main tabs:

COMMAND LINE PARAMETER	EFFECTS ON MAIN TABS
-prodiags	Only displays Information and ProDiags Tabs.
-noprodiags	Displays all tabs except ProDiags.
-healthpage	Only displays Information and Health Page Tabs.
-nohealthpage	Displays all tabs except Health Page.
-modem	Only displays Information, Modem, and Checkout Tabs.
-nomodem	Displays all tabs except Modem and Checkout Tabs.

Table A-1 Command Line Parameter effects on Configuration GUI Tabs

The following files are checked for their existence before allowing a tab to be displayed:

TAB	FILE(S)
Information	None.
ProDiags	/usr/g/insite/ProDiags/bin/pd_Install
Health Page	/usr/g/bin/healthpage
Custom Health Page	/usr/g/config/healthpg.cfg /usr/g/config/healthpg.pm Note, this does not prevent tab from initializing. Alternate means are executed to generate these files.
Modem/Custom Modem/Checkout	/usr/g/insite/bin/installinsite /usr/g/insite/bin/installmodem

Table A-2 Files required for a Tab to appear

ps -ef | grep getty

SGI command string to determine if the port monitor is running

Sample Output:

```
root 15282 1 0 Dec10 ttyf1 0:00 /usr/lib/uucp/uugetty Nt 60 ttyf1 dx_115200
```

grep uugetty /etc/inittab

SGI command to determine the state of the port monitor

Sample Output:

```
#for ports with modems. See the getty(1M), uugetty(1M), init(1M),
t1:23:respawn:/usr/lib/uucp/uugetty -Nt 60 ttyf1 dx_115200 # Port 1 PPP
(2)
```

setReg -r ""

InSite command to see if the modem is alive. Returns OK if success.

licheck

InSite command to check license status of II applications.

Sample Output:

bad.iip_obt: Not Valid: Hash check failed ← Indicates file changed or file not
for that machine

iip_cmesg: Valid

iip_miia: Valid← Indicates License is good

iip_obt: Valid

iip_obt: Not Valid: Expired← indicates license expired, past date limit

May also see a "java.net.UnknownServiceException: no content-type" message.
This message indicates the Web server is not running

Appendix B

InSite Interactive Command Options

SETREG OPTIONS:

Modem Register Access Tool for InSite Interactive Platform

Usage: setReg -p port -r "register" [-s speed] [-d seconds]
setReg -h

-d seconds delay in seconds. This is optional. Sometimes an additional delay is needed to get a response back from the modem. Default is 2 seconds.

-h this helpful output

-p port the port the modem is connected to. Example:
 /dev/cua/a or /dev/ttyf1 .

-r "register" register string to pass to the modem. MUST BE IN DOUBLE QUOTES. May take the following forms:
 -r "S0=1"
 -r "S0=1S2=2S3=4"
 -r "S0=1 S2=2 S3=4"

-s speed serial port speed. This is optional. Typical speeds are 115200, 57600, 38400, 19200, 9600. Default is 38400.

CONFIGLINK OPTIONS:

usage: ConfigLink [-v] [-p] [-d] [-P]

-v - verbose

-P - append -ppp to hostname (GEMS devenv only)

-p - pre-checkout (answer incoming calls only)

-d - debug (set idle timeouts to 30 sec, log activity)

INSTALLINSITE OPTIONS:

Note: "iipadmin config" should be used to configure insite parameters instead of installinsite
usage: installinsite [Options]

Options:

[].....InSite Interactive Installation with GUI Interface

[-debug].....Turn Debugging ON

[-dn].....Report Current Dialout Number

[-dp].....Report Current Dialout Prefix

[-dp DialPre].....Set Current Dialout Number

[-healthpg].....Set up Health Page only with GUI Interface

[-ip].....Report Scanner IP Address

[-ld].....List Network Devices

```
[-ldp].....Report Available Dial Prefixes
[-modem].....Setup modem & InSite Checkout only with GUI Interface
[-nohealthpg].....Setup using the GUI Interface without Health Page
[-nomodem].....Setup using GUI Interface w/o modem & Insite Checkout
[-noprodiags].....Set up using the GUI Interface without Proactive Diags
[-p].....Report Current Phone Number Info
[-prodiags].....Set up Proactive Diags only with GUI Interface
[-tp].....Report Current Line Type (TONE) or (PULSE)
[-tp t|p].....Set Current Line Type (T)one or (P)ulse
{-tty}.....tty InSite Interactive Installation
```

INSTALLMODEM OPTIONS:

Note: "iipadmin config" **should be used to configure modem parameters instead of** installmodem
Usage: installmodem [Options]

Options:

```
[].....Modem Installation
[-debug].....Turn Debugging ON
[-h].....Remote Boot Modem Hangup
[-i].....Current Configuration Inquiry
[-l].....List Available Modem Types
[-ld].....List Modem Menu Descriptions
[-m ModemType].....Install Specific Modem Type
[-os on|off].....Setup OS for Modem Dial In/Out
[-p].....Program CPU Board NVRAM Only
[-q].....Query Modem Registers
[-r].....Reinitialize Modem Registers
[-zsmon ].....Check <zsmon> Port Monitor
```

IIPADMIN OPTIONS:

Administration tool for InSite Interactive

Usage:

```
iipadmin [-r root] [-f file] [-e extdir] [-m dir] [-k] install
iipadmin [-r root] -b file {backup, restore}
iipadmin [-r root] command
iipadmin [-r root] [-m dir] M-command
iipadmin -h
-e extdir specify an extensions directory
-b file "IIP backup" file
-f file install "file" as $INSITE_HOME/.insiterc.local this file
                                must contain the "setenv INSITE_HOME ..."
                                directive
-r root use the InSite Interactive root directory(should end with
"insite")
```

```
-k      keep (do not /bin/rm) existing $INSITE_HOME directory
        during installation
-m dir  specify the directory that contains the class-M
        distribution
-h      this helpful output

commands:
  backup  backup IIP files to standard out
  config  configure to set modem / PPP
  install  initial install and configure to set site parameters
           if -m given, this will also do M-install
  restore  restore previously backed-up files
  start    start all deamons
  stop     stop all deamons

M-commands:
  M-install    install class M package only
  M-uninstall  uninstall class M package only
```

IIPADMIN-BASIC OPTIONS:

Administration tool for iip-pppd

```
Usage:    iipadmin-basic {install | start | stop | uninstall}
install   initial install of iip-pppd to set site parameters
start     start the iip-pppd
stop      stop the iip-pppd
uninstall uninstall the insite directory and associated files
```

IIPADMIN-SERVER OPTIONS:

Administration tool for iip-httpd

```
Usage: iipadmin-server [-n nice]{install | start | stop | halt | uninstall}
-n      set a nice value to lower the process priority
install  initial install of iip-httpd to set site parameters
uninstall unconfigure iip-httpd to original distributed state
start    start the iip-httpd
stop     stop the iip-httpd using SIGTERM
halt     force exit the iip-httpd using SIGKILL
```

GZIP OPTIONS:

gzip 1.2.4 (18 Aug 93)

usage: gzip [-cdfhlLnNtvV19] [-S suffix] [file ...]

```
-c --stdout      write on standard output, keep original files unchanged
-d --decompress  decompress
-f --force       force overwrite of output file and compress links
-h --help       give this help
-l --list        list compressed file contents
-L --license     display software license
```

-n --no-name	do not save or restore the original name and time stamp
-N --name	save or restore the original name and time stamp
-q --quiet	suppress all warnings
-S .suf --suffix .suf	use suffix .suf on compressed files
-t --test	test compressed file integrity
-v --verbose	verbose mode
-V --version	display version number
-1 --fast	compress faster
-9 --best	compress better
file...	files to (de)compress. If none given, use standard input.

VIEWLOG OPTIONS:

Usage: [-h] [filename]

-h prints this help page of .

filename name of the log file to view

allows an ascii terminal to view a log file. If no filename is passed in, the gesyslog is viewed.

PPPDRESTART OPTIONS:

Usage: pppdRestart

Used to restart the ppp daemon

PD_INSTALL OPTIONS:

Usage: pd_Install [Options]

Options:

[].....Normal ProDiags Installation

[-auto].....Auto ProDiags Install (No User Questions)

[-debug].....Turn Debugging ON

pd_Install: '-h' is not a valid command line option

GATTENTION OPTIONS:

usage: gattention ["text message"]

This command creates a pop-up Attention! window.

Example: gattention "Test Message"

PRODIAGS OPTIONS:

Usage: prodiags [Options]

The user interface to Proactive Diagnostics

Options:

[].....Start Normal User Interface

[-e].....Start at 'Execute a Task' Menu

[-p].....Start at 'Change Phone Number' Menu

[-page size].....Adjust Display Page to 'size' Lines

[-s].....Start at 'Scheduler' Menu

[-sm].....Start at 'Modify Task Schedule' Menu
[-u].....Start at 'Utilities' Menu
[-vl].....Start at 'View Log' Menu
[-vr].....Start at 'View Results' Menu

LESS OPTIONS:

Usage: less [filename]

SUMMARY OF COMMANDS

Commands marked with * may be preceded by a number, N.

Notes in parentheses indicate the behavior if N is given.

h H Display this help.
q :q :Q ZZ Exit.
e ^E j ^N CR * Forward one line (or N lines).
y ^Y k ^K ^P * Backward one line (or N lines).
f ^F ^V SPACE * Forward one window (or N lines).
b ^B ESC-v * Backward one window (or N lines).
z * Forward one window (and set window to N).
w * Backward one window (and set window to N).
d ^D * Forward one half-window (and set half-window to N).
u ^U * Backward one half-window (and set half-window to N).
F Forward forever; like "tail -f".
r ^R ^L Repaint screen.
R Repaint screen, discarding buffered input.

Default "window" is the screen height.

Default "half-window" is half of the screen height.

/pattern * Search forward for (N-th) matching line.
?pattern * Search backward for (N-th) matching line.
n * Repeat previous search (for N-th occurrence).
N * Repeat previous search in reverse direction.
ESC-n * Repeat previous search, spanning files.
ESC-N * Repeat previous search, reverse dir. & spanning files.
ESC-u Undo (toggle) search highlighting.

Search patterns may be modified by one or more of:

! search for NON-matching lines.

* search multiple files.

@ start search at first file (for /) or last file (for ?).

g < ESC-< * Go to first line in file (or line N).
G > ESC-> * Go to last line in file (or line N).
p % * Go to beginning of file (or N percent into file).
{ * Go to the } matching the (N-th) { in the top line.

}	* Go to the { matching the (N-th) } in the bottom line.
(	* Go to the) matching the (N-th) (in the top line.
)	* Go to the (matching the (N-th)) in the bottom line.
[	* Go to the] matching the (N-th) [in the top line.
]	* Go to the [matching the (N-th)] in the bottom line.
ESC-^F <c1> <c2>	* Go to the c1 matching the (N-th) c2 in the top line
ESC-^B <c1> <c2>	* Go to the c2 matching the (N-th) c1 in bottom line.
m<letter>	Mark the current position with <letter>.
'<letter>	Go to a previously marked position.
''	Go to the previous position.
^X^X	Same as '.
:e [file]	Examine a new file.
^X^V	Same as :e.
:n	* Examine the (N-th) next file from the command line.
:p	* Examine the (N-th) previous file from the command line.
:x	* Examine the first (or N-th) file from the command line.
= ^G :f	Print current file name.
V	Print version number of "less".
-<flag>	Toggle a command line flag [see FLAGS below].
_<flag>	Display the setting of a command line flag.
+cmd	Execute the less cmd each time a new file is examined.
!command	Passes the command to \$SHELL to be executed.
X_Xcommand	Pipe file between current pos & mark X_X to shell command.
v	Edit the current file with \$VISUAL or \$EDITOR.

FLAGS

Most flags may be changed either on the command line,
or from within less by using the - command.

-?	Display help (from command line).
-a	Forward search skips current screen.
-b [N]	Number of buffers.
-B	Don't automatically allocate buffers for pipes.
-c -C	Repaint by scrolling/clearing.
-d	Dumb terminal.
-e -E	Quit at end of file.
-f	Force open non-regular files.
-g	Don't highlight matches for previous search pattern.
-G	Highlight ALL matches for previous search pattern.
-h [N]	Backward scroll limit.
-i	Ignore case in searches.
-I	Ignore case in searches and in search patterns.
-j [N]	Screen position of target lines.
-k [file]	Use a lesskey file.
-m -M	Set prompt style.

-n -N	Use line numbers.
-o [file]	Log file.
-O [file]	Log file (unconditionally overwrite).
-p [pattern]	Start at pattern (from command line).
-P [prompt]	Define new prompt.
-q -Q	Quiet the terminal bell.
-r	Output "raw" control characters.
-s	Squeeze multiple blank lines.
-S	Chop long lines.
-t [tag]	Find a tag.
-T [tagsfile]	Use an alternate tags file.
-u -U	Change handling of backspaces.
-V	Display the version number of "less".
-w	Display ~ for lines after end-of-file.
-x [N]	Set tab stops.
-X	Don't use termcap init/deinit strings.
-y [N]	Forward scroll limit.
-z [N]	Set size of window.

Appendix C

IIP Configuration Tool Troubleshooting Text

Note: Causes and Solutions are NOT in order of frequency. Cause 1 is resolved by Solution 1.)

Section 1.0 PROBLEM: IIP Conf. GUI Hangs When Configuring Port Monitor (i.e., Serial Port)

1.1 CAUSES:

- 1.) Modem is causing a serial port hang.
- 2.) Orphan ppp process is using the serial port.
- 3.) Bad Serial Port.
- 4.) Bad Modem.

1.2 SOLUTION:

- 1.) Turn Modem power off for 5 seconds, turn modem back on, retry to configure the modem.
- 2.) Type: \pppdRestart kill ENTER to stop all ppp processes.
- 3.) Replace bad Serial Port.
- 4.) Replace bad Modem.

Section 2.0 PROBLEM: Check Hardware message During Modem Configuration.

2.1 CAUSES:

- 1.) Modem is connected to a different serial port.
- 2.) Bad modem to CPU serial cable.
- 3.) Modem is not turned on.
- 4.) Modem is off hook (i.e., connected to OnLine Center).
- 5.) Bad Modem.
- 6.) TTY port not configured correctly.

2.2 SOLUTIONS:

- 1.) Connect the modem to the correct serial port.
- 2.) Replace the serial cable between the modem and the CPU.
- 3.) Turn on the Modem.

- 4.) Press the button on the front of the modem to disconnect.
- 5.) Replace the modem.
- 6.) Start the IIP Configuration Tool, and select APPLY on the MODEM tab.

Section 3.0 PROBLEM: Check Hardware message During Configuration of Motorola Modems.

3.1 CAUSES:

Modem set to an unusual factory settings or went through a modem configuration as an USRobotics modem.

3.2 SOLUTIONS:

Reset DCD to Normal before configuring the modem.

- 1.) Behind the Motorola modem front panel, select the Return key (diagonal arrow) until you see "Disconnect T/D?"
- 2.) Select the down arrow until you see Terminal Opts
- 3.) Select the right arrow button until you see DCD= 'value' this value must be "Normal", otherwise, Select the down button until the display reads, DCD = Normal.
- 4.) Press the enter button.
- 5.) Select the Return key (diagonal arrow) until you see "Disconnect T/D?"
- 6.) Reconfigure the modem using the IIP Configuraiton GUI.

Section 4.0 PROBLEM: InSite Cannot Connect to the site.

4.1 CAUSES:

- 1.) Modem is not turned on.
- 2.) Phone Line to modem is disconnected or in wrong jack.
- 3.) Modem to CPU serial cable is not connected.
- 4.) Bad modem to CPU serial cable
- 5.) InSite phone number has changed or been disconnected.
- 6.) The insite user password has changed.
- 7.) CPU serial port is bad.
- 8.) PPP daemon is not running.

4.2 SOLUTIONS:

- 1.) Turn on the modem.
- 2.) Connect the phone line to the port labeled \Jack\ .

- 3.) Connect the cable between the Modem and the CPU.
- 4.) Replace the serial cable
- 5.) Verify the Phone Line used by InSite is functional.
- 6.) Call the OLC to update the insite password in the database.
- 7.) Replace the CPU serial port used by the modem.
- 8.) Run the command \pppdRestart\ as root.

Section 5.0 PROBLEM: Dial out to AutoSC or OnLine Center Fails. Dial in is OKAY.

5.1 CAUSES:

- 1.) Wrong Dial out Prefix.
- 2.) Wrong Dial out Phone Number.
- 3.) PPP routing is incorrect or missing.

5.2 SOLUTIONS:

- 1.) Change Dial out Prefix in the IIP Configuration Tool.
- 2.) Call the OLC for an InSite Checkout.
- 3.) Call the OLC for an InSite Checkout.

Section 6.0 PROBLEM: Web Browser not functioning properly.

6.1 CAUSES:

- 1.) Web browser not responding to requests or actions.
- 2.) Web browser at an unknown location and can't get to home page.
- 3.) Web browser distorts pages.

6.2 SOLUTIONS:

For all these cases, close the current Web Browser and start a new browser with the command:
`insite_browser`

Section 7.0 PROBLEM: Web Server not functioning properly.

7.1 CAUSES:

- 1.) New files added under the ~insite/classes/com directory tree.
- 2.) Too many iip-http processes running.
- 3.) Web server cannot find pages on system.
- 4.) Abandoned httpd processes.

7.2 SOLUTIONS:

For all these casues, in the ~insite/server directory, run the command:
`\iipadmin stop halt start\`

Section 8.0 PROBLEM: License error when trying to run an InSite Interactive Application.

8.1 CAUSES:

- 1.) License for an Application does not exist.
- 2.) License for an Application is invalid.
- 3.) License for an Application has expired.

8.2 SOLUTIONS:

For all these casues, call the OLC to refresh applicable InSite Interactive licenses.

Section 9.0 IMPORTANT LOGS TO VIEW FOR INSITE PROBLEMS:

Note: Use \viewlog\ found in the ~insite/bin to view these logs.

Error Log Name	Error Log Purpose
-----	-----
\$IIP_SESSION_LOG	General InSite error log
~insite/logfiles/configdebug.log	Config Tool debug log
~insite/server/logs/access_log	Web server access log
~insite/server/logs/error_log	Web server error log
/var/adm/SYSLOG or /var/adm/pppd.log	ppp error log

Section 10.0

IMPORTANT COMMANDS FOR INSITE INTERACTIVE:

Command	Command Purpose
-----	-----
iiadmin config	start InSite Configuration GUI
iiadmin config -debug	start InSite Config GUI and create debug log
iiadmin config -tty	InSite Configuration in tty mode
iiadmin stop start	Stop then restart Web server
insite_browser	start an InSite Interactive browser
viewlog <log name>	view a log with the viewlog utility

Appendix D

Example of IIP Configuration Files

IIP CONFIGURATION FILES

Note: **BOLD** text indicates site specific information, **BOLD Italic** indicates site specific information supplied by the OLC in the `sclink.cfg` file.

FILE: `~insite/.insiteINFO`

```
INS_Connection Type="Modem"
INS_ModemType="ModemUSRCourier"
INS_PhoneNumber="8005708878"
INS_DialPrefix="9,1"
INS_ToneOrPulse="T"
INS_Country="Default - All Others"
INS_ModemString="&F1 X4 &D0 E0 T M0 Q2 S0=1 S7=90 "
INS_TtyPort="/dev/ttyf1"
INS_TtySpeed="38400"
INS_Gateway IP ="38400"
INS_Gateway MASK ="38400"
```

FILE: `~insite/.insiterc.local`

```
#-----
# Copyright (c) 1998 The General Electric Company
#-----
# This file contains site-specific Insite Interactive Platform (IIP)
# parameter settings to be used during installation, configuration and
# execution.

# The format of this file is a /bin/csh source-able file.

# Settings in this file (.insiterc.local) will override those set by the
# ~insite/.insiterc file.

# Do not source this file directly! This file is automatically sourced
# from ~insite/.insiterc which MUST be used to source this file.

# NOTE: VERIFY ALL CSH EXECUTABLE COMMAND STRINGS ARE CORRECT BEFORE
# RELEASE!

#-----
# Uncomment and/or set variables below, as needed.
#-----
```

```
# Additional .cshrc files your modality may need Insite Interactive to source
# -----
# source ~_YOUR_USER_HERE_/.cshrc
#=====
#
#           M A N D A T O R Y       S E C T I O N
#=====
# INSITE_HOME must be defined as a full path name, ending with "/insite".
# -----

setenv INSITE_HOME /usr/g/insite
# Set file name for modality specific information
# -----
setenv IIP_MODALITY_INFO /usr/g/config/INFO

# Set Configuration directory
# -----
setenv IIP_CONFIG_DIR /usr/g/config

# Set Health Page executable directory, The directory where the healthpage
# executable can be found. Normally $INSITE_BIN
# -----
setenv IIP_HEALTHPAGE_DIR /usr/g/bin

# Set HealthPage Config Files directory, The directory where the healthpage
# configuration files can be found. Normally, the same as $IIP_CONFIG_DIR
# -----
setenv IIP_HP_CONFIG_DIR /usr/g/config

# Set list_select_ex executable directory, The directory where the
# list_select_ex executable can be found. Normally $INSITE_BIN
# -----
setenv IIP_LIST_SELECT_DIR /usr/g/insite/bin

# csh executable Command string to set Hospital Name
# -----
set IIP_HOSP=`grep hospitalName /usr/g/config/Suite.cfg | awk -F= '{print $2}'`
setenv IIP_HOSPITALNAME "$IIP_HOSP"

# Path and Filename of the systems password file.
# (used to check for changed passwords)
# -----
setenv IIP_PASSWD_FILE /etc/passwd

#=====
#
#           O P T I O N A L       S E C T I O N
#=====
# IIP_II_MODALITY should be one of: CT, MR, XR, US, II (Insite
# Interactive), ...
```



```
# -----
setenv IIP_II_MODALITY CT

# IIP_SERVLET_PORT defines the port for HTTP servlets.
# Default is port 8182 and does not need to be set if using the default.
# -----
# setenv IIP_SERVLET_PORT _not_8182_

# IIP_SERVER_NICE defines the nice value for the web server
# Default is no NICE value given
# (Can also be set using -nice option on the iipadmin-server command line)
# -----
setenv IIP_SERVER_NICE 10

# Additional paths your modality may need added to LD_LIBRARY_PATH.
# This is a colon seperated list and must include the default definition.
# For IRIX 6.4 (and beyond?) LD_LIBRARYN32_PATH and LD_LIBRARY64_PATH
# must also be defined iff LD_LIBRARY_PATH is defined.
# -----
setenv LD_LIBRARY_PATH /opt/X11R5/lib:/usr/lib/Motif:/usr/g/ctuser/lib:/usr/lib:
setenv LD_LIBRARYN32_PATH ${LD_LIBRARYN32_PATH}:/usr/g/ctuser/lib32:/usr/lib32:/
lib32:/usr/g/ctuser/lib32
setenv LD_LIBRARY64_PATH ${LD_LIBRARY64_PATH}:/usr/lib64:/lib64

# Path and FileName of the InSite Session Log
# -----
# setenv IIP_SESSION_LOG _YOUR_SESSION_LOGNAME_
# setenv IIP_SESSION_LOG /usr/g/service/state/ssw.srv_activity.ref
# Path and Filename of the default log to be used by viewlog
# NOTE: include both lines (set HOSTNAME and setenv IIP_VIEWLOG_FILENAME).
# -----
# setenv IIP_VIEWLOG_FILENAME _YOUR_LOG_FILENAME_HERE_
set HOSTNAME=`uname -n`
setenv IIP_VIEWLOG_FILENAME /usr/g/service/log/gesys_${HOSTNAME}.log

# csh executable Command string to notify the user InSite is logged in
# -----
# setenv IIP_NOTIFY_II_LOGIN "insite_msg 0 >&! /dev/null &"
setenv IIP_NOTIFY_II_LOGIN "/usr/g/bin/ctiiputil -login"

# csh executable Command string to notify the user InSite is logged out
# -----
# setenv IIP_NOTIFY_II_LOGOUT "insite_msg 1 >&! /dev/null &"
setenv IIP_NOTIFY_II_LOGOUT "/usr/g/bin/ctiiputil -logout"

# csh executable Command string to Set Software Revision
# -----
setenv IIP_SOFTWARE_REV `showprods | grep "Helios_OC_Product " | cut -d" " -f11`
```

```
# For Perl, if not using the IIP available Perl distribution,
#         set IIP_PERL_BIN to the full pathname to its binary and

# Defaults are already set for using the IIP Perl distribution:
#   IIP_PERL_BIN = $INSITE_HOME/perl/bin/perl
# -----
# setenv IIP_PERL_BIN /usr/sbin/perl          ... for example
#####
# NOTE: Let IIP use it's own perl version - DON'T DEFINE IT...
#####
# csh executable Command string to obtain the Service ID from modality
# a configuration file
# -----
# setenv IIP_SERVICE_ID `grep ServiceID $IIP_CONFIG_DIR/host.cfg|awk -F=
#{print $2}`
#=====
#                               End of .insiterc.local file
#=====
```

FILE: ~insite/sclink.cfg

```
LinkID=336300
LinkSecret=dadhcffc356
ProductPhone=8w4145216680
ProductIP=172.23.184.3/255.255.255.255
CenterPortmasterName=olc-pm1
USN=00000GSAO3
ServiceID=GSAO3
MLN=9999
IIWebIP=3.7.192.187
IIWebPort=8030
IIWebName=olcwebd
CenterPhone=8005708878
GESubnet=3.245.60.0/255.255.252.0
GESubnet=3.245.172.0/255.255.252.0
GESubnet=3.245.250.0/255.255.252.0
GESubnet=3.245.68.0/255.255.252.0
GESubnet=3.87.168.0/255.255.252.0
GESubnet=3.70.152.0/255.255.252.0
GESubnet=3.57.88.0/255.255.252.0
GESubnet=3.45.16.0/255.255.252.0
GESubnet=3.245.32.0/255.255.252.0
GESubnet=3.7.188.0/255.255.252.0
GESubnet=3.7.192.0/255.255.252.0
GESubnet=3.87.88.0/255.255.252.0
```

GESubnet=3.87.16.0/255.255.252.0
ASCIP=3.87.88.3
ASCIP=3.87.88.2
ASCIP=3.87.88.1
CenterPortmasterIP=3.87.88.221
Dial Prefix = 9,1

SGI PPP FILES

Note: **BOLD text indicates site specific information, BOLD Italic indicates site specific information supplied by the OLC in the `sclink.cfg` file.**

FILE: `/etc/ppp.conf`

```
# FILE: /etc/ppp.conf
# NOTE: This file was automatically generated by:
#       /export/home/insite/ConfigLink on Tue Dec 15 16:04:16 CST 1998
#       Modifications made to this file will
#       be lost the next time ConfigLink runs.
```

```
olc-pm1 quiet remotehost=3.87.88.221
        localhost=172.23.184.3
        send_passwd=dadhcffc356
        send_name=336300
        send_chap
        inactive_timeout=180
        -mp -ccp
        debug=1
```

```
Pinsite in remotehost=3.87.88.221
        localhost=172.23.184.3
        recv_passwd=dadhcffc356
        send_name=336300
        send_chap
        rem_sysname=olc-pm1
        -mp -ccp
```

```
_INCOMING reconfigure
```

FILE: `/etc/uucp/Systems`

```
#ident "$Revision: 1.14 $"
```

```
# Entries have this format:
```

```
#Machine-Name Time Type Class Phone Login
```

```
# Machine-Name node name of the remote machine
# Time          day-of-week and time-of-day when you may call
#               (e.g., MoTuTh0800-1700). Use "Any" for any day.
#               Use "Never" for machines that poll you, but that
#               you never call directly.
# Type          device type
# Class         transfer speed
# Phone         phone number (for autodialers) or token (for
```

```
# data switches)
# Login login sequence is composed of fields and subfields
# in the format "[expect send] ...". The expect field
# may have subfields in the format "expect[-send-expect]".
# The special characters in the "expect send" pairs
# is documented in the Dialers file.

# Example:
# cuuxb Any ACU 1200 chicago8101242 in:--in: nuucp word: panzer

# This example uses one of the example lines in Devices, with a Hayes 2400:
# sgi Any ACU 2400 14155551212 "" @\r\c ogin:--ogin:-\b\d-ogin:--ogin: @nuucp
# This example uses UUCP/TCP and the 't' protocol. The 'e' protocol
# could be used instead.
# rhost Any TCP,t Any rhost.foo.com ogin: Urhost assword: guessit
# This example is for an ISDN connection for a PPP link.
# ishost Any ISDN Any "" "" ISDNCALL[64]5551212 CONNECTED
# Add Definition for insite Modem
    olc-pm1 Any ACU 38400 T9,18005708878
```

FILE: /etc/uucp/Devices

```
# $Revision: 1.16 $

# All lines must have at least 5 fields. Use '-' for unused fields.

# The Devices file is used with the Dialers file.
# The first field of a line in the Dialers file is the name.

# Types that appear in the 5th field must either name built-in
# functions (801 or 212) or name lines in the Dialers file.

# All modem lines should be of the form:
# Name device null speed 212 x dialer

# There can be one or more (dialer, arg) pairs following the 4th field.
# The "arg" fields (6th, 8th, and so on) are "telephone numbers," optionally
# using the \D or \T escape sequences.
# If missing, the 6th field, the arg for the first dialer, is assumed to be
# \T. Subsequent telephone numbers are assumed to be \D if missing.
# \T and \D have the same meanings in the arg strings here as they do
# in the Dialers file, the telephone number from the Systems file,
# translated thru the Dialcodes file if \T.
# Blank lines and lines that begin with a <space>, <tab>, or # are ignored.
# Protocols can be specified as a comma-subfield of the device type
# either in the Devices file (where device type is field 1)
# or in the Systems file (where it is field 3).
```

```
# --Standard modem line
# ACU ttym2 null 1200 212 x hayes24

# --A direct line so 'cu -lttyd2' will work
# Direct ttyd2 - 9600 direct

# --A Hayes 2400 modem, at 2400 bps
#           The phone number
# ACU ttym4 null 2400 212 x hayes24

# --A direct connection to a system
# systemx ttyd2 - Any direct
#
# --where the Systems file entry looks like:
# systemx Any systemx 9600 unused  "" in:-\r\d-in: nuucp word: nuucp
#      (The third field in Systems matches the first field in Devices)

# For UUCP over TCP
#      The 6th field is the port number.
TCP - - Any TCP uucp

# For PPP over ISDN
ISDN isdn/modem_b1 - Any direct
ISDN isdn/modem_b2 - Any direct

#ACU ttym2 null 1200 212 x hayes24
#ACU ttym2 null 2400 212 x hayes24
#Direct ttyd2 - Any direct

#ACU ttym3 null 1200 212 x hayes24
#ACU ttym3 null 2400 212 x hayes24
#Direct ttyd3 - Any direct

# Typical TB+ entry.  Notice the use of ttyf instead of ttym.
#ACU ttyf4 null 19200 212 x telebit
#ACUPEP ttyf4 null 19200 212 x telebitpep
#ACUCOM ttyf4 null 19200 212 x telebitcom
#ACUSLIP ttyf4 null 19200 212 x teleslip
#Direct ttyd4 - Any direct

# 1st optional SIO board

#ACU ttym5 null 1200 212 x hayes24
#ACU ttym5 null 2400 212 x hayes24
#Direct ttyd5 - Any direct

#ACU ttym6 null 1200 212 x hayes24
#ACU ttym6 null 2400 212 x hayes24
#Direct ttyd6 - Any direct
```

```
#ACU ttym7 null 1200 212 x hayes24
#ACU ttym7 null 2400 212 x hayes24
#Direct ttyd7 - Any direct

#ACU ttym8 null 1200 212 x hayes24
#ACU ttym8 null 2400 212 x hayes24
#Direct ttyd8 - Any direct

#ACU ttym9 null 1200 212 x hayes24
#ACU ttym9 null 2400 212 x hayes24
#Direct ttyd9 - Any direct

#ACU ttym10 null 1200 212 x hayes24
#ACU ttym10 null 2400 212 x hayes24
#Direct ttyd10 - Any direct

#ACU ttym11 null 1200 212 x hayes24
#ACU ttym11 null 2400 212 x hayes24
#Direct ttyd11 - Any direct

#ACU ttym12 null 1200 212 x hayes24
#ACU ttym12 null 2400 212 x hayes24
#Direct ttyd12 - Any direct
ACU /dev/ttyf1 null 38400 212 x USRobotics
```

FILE: /etc/uucp/Dialers

```
# ident "$Revision: 1.40 $"
# Each caller type that appears in the Devices file (5th field)
# should appear in this file except for the built in callers, 212 and 801.
# Each line consists of three parts:
# 1. the name of the caller,
# 2. the translation table for the phone number to translate from
# the 801 codes (=) to the code for the particular device,
# 3. a chat script with the same format and meaning as the login scripts
# that appear in the Systems file.
#
# Meaning of some of the escape characters:
# \p - pause (approximately 1/4-1/2 second delay)
# \d - delay (2 seconds)
# \D - phone number/token
# \T - phone number with Dialcodes and character translation
# \N - null byte
# \K - insert a BREAK
# \E - turn on echo checking (for slow devices)
# \M - no modem control, set CLOCAL (ignore DCD)
```

```
# \m - modem control, clear CLOCAL (wait for DCD on open, hangup on DCD
loss)
# \e - turn off echo checking
# \r - carriage return
# \c - no new-line
# \n - send new-line
# \s - send space
# \t - send tab
# \b - send backspace
# \\ - send \
# \nnn - send octal number
# ~nn - change the timeout from 45 to nn seconds for waiting for this
string.
#      This setting must be at the end of expect string.

# The expect-send sequence 'ABORT BUSY' cause chatting to be aborted
# if the string 'BUSY' is received in subsequent chatting. There can be
# as many as 5 'ABORT str' pairs. The pair 'ABORT ""' turns off all
# of the abort strings. All of the abort strings are turned off at
# the start of chatting; abort strings set in the Dialers file have no
# effect when chatting is restarted in the Systems file.

# Blank lines and lines that begin with a <space>, <tab>, or # are ignored.

#      simple, direct connection, and ignore DCD
direct      "" "" \M\c

#      direct connection using uugetty with the -r option on both ends.
uudirect "" "" \d\r\d in:--in:

#      Rixon Intelligent Modem -- modem should be set up in the Rixon
#      mode and not the Hayes mode.
rixon =&-%  "" \d\r\r\d $ s9\c )-W\r\ds9\c-) s\c : \T\r\c $ 9\c LINE

penril=W-P  "" \d > Q\c : \d- > s\p9\c )-W\p\r\ds\p9\c-) y\c : \E\TP > 9\c
OK
ventel=&-%  "" \d\r\p\r\c $ <K\T%\r>\c ONLINE!
vadic =K-K  "" \d\005\p *-005\p-*005\p-* D\p BER? \E\T\e \r\c LINE

#####
#      Hayes Smartmodem -- modem should be set with the configuration
#      switches as follows:
#          S1 - UPS2 - UP   S3 - DOWNS4 - UP
#          S5 - UPS6 - DOWNS7 - ?S8 - DOWN
hayes =,-,  "" \dAT\r\c OK \EATDT\T\r\c CONNECT~90
```



```
# Hayes 2400, on a Touch-Tone line
# This entry works on some 'Hayes compatible' modems.
# This assumes the modem has been programmed by /etc/uucp/fix-hayes
#
hayes24=W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE ""
\dATelq0&d3s2=128L0\r\c OK-ATelq0&d3s0=0s2=128L0\r\c-OK atdt\T\r\c
CONNECT~90

# Tell uugetty how to ready a Hayes 2400 for incoming calls
# This assumes the modem has been programmed by /etc/uucp/fix-hayes
# "Security" suggests turning off '+++'.
# Some 'Hayes compatible' modems do not really reset when the call
# is finished, but the '&d3' helps them.
# It keeps the speaker muted.
hayes24in=W-, "" \dAT&d3s0=1s2=128L0\r\c OK-AT&d3s0=1s2=128L0\r\c-OK

# Dial using a Hayes ACCURA
# Try hard to turn off the Hayes Patented escape
hayes14=W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE ""
\pATs2=128s12=255s0=0\r\c OK\r~2-ATs2=128s12=255s0=0\r\c-OK\r~2
ATdt\T\r\c CONNECT~90

# incoming chatting for uugetty
hayes14in=W-, "" \pATs2=128s12=255s0=1\r\c OK\r~2-
ATs2=128s12=255s0=1\r\c-OK\r~2

#####
# Telebit Trailblazer
# These assume the modem has been programmed by /etc/uucp/fix-telebit
#
# You want to turn off compression while sending pre-compressed.

# UUCP
# TB+, T2000, or T2500
# Turn off compression, since with pre-compressed data that is faster.
telebit =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE ""
\dAT\r\c OK\r~2-AT\r\c-OK\r~2 ATs0=0s110=0\r\c OK\r ATdtw\T\r\c
CONNECT~90
#
# A T1600. A T2000 or T2500 would be faster.
t16 =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE "" \dAT\r\c
OK\r~2-AT\r\c-OK\r~2 ATdtw\T\r\c CONNECT~90
#
# Using a TB, TB+, T2000, or T2500 for UUCP, call a machine which wants
# things compressed, and may offer PEP last
telebitcom=W-,ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE "" \dAT\r\c
OK\r~2-AT\r\c-OK\r~2 ATs0=0s50=255s110=1\r\c OK\r ATdtw\T\r\c CONNECT~90
#
```

```
# Using a TB, TB+, T2000, or T2500, for UUCP call a machine which
presents
# the PEP tones last, so insist on PEP, but no compression.
telebitpep=W-,ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE "" \dAT\r\c
OK\r~2-AT\r\c-OK\r~2 ATs0=0s50=255s7=60s110=0\r\c OK\r ATdtw\T\r\c
CONNECT~90

# QBlazer: SLIP and UUCP are the same since it does not spoof UUCP
tqbb =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE "" \dAT\r\c
OK\r~2-AT\r\c-OK\r~2 ATs0=0\r\c OK\r ATdtw\T\r\c CONNECT~90

# SLIP over a TB+: compression on, UUCP off, and insist on PEP.
# You may wish to turn off compression to improve latency.
teleslip=W-,ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE "" \dAT\r\c
OK\r~2-AT\r\c-OK\r~2 ATs0=0s50=255s110=1s111=0\r\c OK\r ATdtw\T\r\c
CONNECT~90

#
# SLIP over a T2500, T1600, or T3000: UUCP off
# The T2500 entries should be used with T1000's
t25slip=W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE "" \dAT\r\c
OK\r~2-AT\r\c-OK\r~2 ATs0=0s111=0\r\c OK\r ATdtw\T\r\c CONNECT~90
t16slip=W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE "" \dAT\r\c
OK\r~2-AT\r\c-OK\r~2 ATs0=0s111=0\r\c OK\r ATdtw\T\r\c CONNECT~90
t30slip=W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE "" \dAT\r\c
OK\r~2-AT\r\c-OK\r~2 ATs0=0s111=0\r\c OK\r ATdtw\T\r\c CONNECT~90

# incoming chatting for uugetty
# Tell uugetty about a TB+
telebitin=W-, "" \dAT\r\c OK\r~2-AT\r\c-OK\r~2 ATs0=1\r\c OK\r
#
# Tell uugetty about a T2500 ready to do SLIP: PEP last
t25in =W-, "" \dAT\r\c OK\r~2-AT\r\c-OK\r~2 ATs92=1s0=1\r\c OK\r
#
# Tell uugetty about a T1600, T3000, or QBlazer
t16in =W-, "" \dAT\r\c OK\r~2-AT\r\c-OK\r~2 ATs0=1\r\c OK\r
t30in =W-, "" \dAT\r\c OK\r~2-AT\r\c-OK\r~2 ATs0=1\r\c OK\r
tqbin =W-, "" \dAT\r\c OK\r~2-AT\r\c-OK\r~2 ATs0=1\r\c OK\r

#####
# Digicom Systems, Inc.
# 9624LE+ or Scout+
#
# These assume the modem has been programmed by /etc/uucp/fix-dsi

dsi960=W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE ""
\pATs2=128s0=0\r\c OK\r~2-ATs2=128s0=0\r\c-OK\r~2 ATdt\T\r\c CONNECT~90
```

```
# incoming chatting for uugetty
dsiin =W-, "" \pATs2=128s0=1\r\c OK\r~2-ATs2=128s0=1\r\c-OK\r~2

#####
# ZyXEL
#
#     These assume the modem has been programmed by /etc/uucp/fix-zyxel

# U-1496
zy1496=W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE ""
\d\r\pATs2=128s0=0\r\c OK\r~2-\rATs2=128s0=0\r\c-OK\r~2 ATdtw\T\r\c
CONNECT~90

# incoming chatting for uugetty, including strings to autobaud
zyin =W-, "" \d\r\pAT\r\c OK\r~2-\rAT\r\c-OK\r~2 ATS2=128s0=1\r\c OK\r

#####
# Intel
#
#     These assume the modem has been programmed by /etc/uucp/fix-intel

# Intel 14.4EX
intel14=W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE ""
\dATs2=128s0=0\r\c OK\r~2-\rATs2=128s0=0\r\c-OK\r~2 ATdt\T\r\c
CONNECT~90

# incoming chatting for uugetty
intelin=W-, "" \dAT\r\c OK\r~2-\rAT\r\c-OK\r~2 ATS2=128s0=1\r\c OK\r

#####
# DALLAS FAX
#
#     These assume the modem has been programmed by /etc/uucp/fix-dallas

# Dallas Fax 14.4E Pro Plus
dalpro=W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE ""
\dATs0=0\r\c OK\r~2-\rATs0=0\r\c-OK\r~2 ATdt\T\r\c CONNECT~90

# incoming chatting for uugetty
dalproi=W-, "" \dAT\r\c OK\r~2-\rAT\r\c-OK\r~2 ATs0=1\r\c OK\r

#####
# USRobotics
#
#     These assume the modem has been programmed by /etc/uucp/fix-usr
```

```
# Sportster or Courier
usr =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIAL\sTONE ABORT
NO\sDIALTONE "" \pATs2=128s0=0\r\c OK\r~2-\rATs2=128s0=0\r\c-OK\r~2
ATdt\T\r\c CONNECT~90

# incoming chatting for uugetty
usrin =W-, "" \pAT\r\c OK\r~2-\rAT\r\c-OK\r~2 ATs2=128s0=1\r\c OK\r

#####
# AT&T Paradyne
#
#       These assume the modem has been programmed by /etc/uucp/fix-att

# DataPort 14.4/FX
attdp =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE ""
\pATZ0s0=0\r\c OK\r~2-\rATZ0s0=0-OK\r~2 ATs2=128S79=0X6&C1&D3&Y0\r\c-OK
OK ATdt\T\r\c CONNECT~90

# incoming chatting for uugetty
attin =W-, "" \pATZ0\r\c OK\r~2-\rATZ0\r\c-OK\r~2
ATS0=1S2=128X6&C1&D3&Y0 OK\r

#####
# MultiTech
#       These modems do not like to keep DSR true; so ignore DCD while
programming
#       them.
#
#       These assume the modem has been programmed by /etc/uucp/fix-
multitech

# ZDX
zdx =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE ""
\p\MAT&F0&D3&SF1&S0%CD1s0=0\r\c OK\r~2-AT&F0&D3&SF1&S0%CD1s0=0-OK\r~2
\mAT&E11%E0$F0X4S13=128S36=0S37=0\r\c-OK\r~2 OK ATdt\T\r\c CONNECT~90

# incoming chatting for uugetty
zdxin =W-, "" \p\MAT&F0&D3&SF1&S0%CD1\r\c OK\r~2-
\rAT&F0&D3&SF1&S0%CD1\r\c-OK\r~2 \mAT&E11%E0$F0X4S13=128S36=0S37=0S0=1
OK\r

#####
# look for "CONNECT" for uugetty.  Uugetty can use this to know whether
# to shutdown or to look for the username.  This significantly reduces
# the time the modem is off-hook after a wrong number.
conn =W-, ABORT NO\sCARRIER CONNECT
```

```
#
#       Modem Dialer   : 3COM Modem Types
#       File           : $INSITE_HOME/.pppdialer.3COM
#       Created By      : Mike Suchecki
#       Original Date   : August 25, 1998
#
3COM =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIAL\STONE ABORT
NO\sDIALTONE "" \pAT\r\c OK\r~2-\r\c-OK\r~2 ATdt\T\r\c CONNECT~90
#
#       Modem Dialer   : Generic Modem Types
#       File           : $INSITE_HOME/.pppdialer.Generic
#       Created By      : George Tzortzos
#       Original Date   : October 22, 1997
#
Generic =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIAL\STONE ABORT
NO\sDIALTONE "" \pAT\r\c OK\r~2-\r\c-OK\r~2 ATdt\T\r\c CONNECT~90
#
#       Modem Dialer   : Hayes Modem Types
#       File           : $INSITE_HOME/.pppdialer.Hayes
#       Created By      : George Tzortzos
#       Original Date   : October 20, 1997
#
Hayes_optima =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE ""
\pATs2=128s12=255s0=1\r\c OK\r~2-ATs2=128s12=255s0=1\r\c-OK\r~2
ATdt\T\r\c CONNECT~90
#
#       Modem Dialer   : Motorola Codex Modem Types
#       File           : $INSITE_HOME/.pppdialer.Motorola
#       Created By      : George Tzortzos
#       Original Date   : October 20, 1997
#
Motorola =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIAL\STONE ABORT
NO\sDIALTONE "" \pAT\r\c OK\r~2-\r\c-OK\r~2 ATdt\T\r\c CONNECT~90
#
#       Modem Dialer   : MultiTech Modem Types
#       File           : $INSITE_HOME/.pppdialer.MultiTech
#       Created By      : Mike Suchecki
#       Original Date   : April 2, 1999
#
MultiTech =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIALTONE "" \pAT\r\c
OK\r~2-\r\c-OK\r~2 ATdt\T\r\c CONNECT~90
#
#       Modem Dialer   : US Robotics Modem Types
#       File           : $INSITE_HOME/.pppdialer.USRobotics
#       Created By      : George Tzortzos
#       Original Date   : October 22, 1997
```

#

USRobotics =W-, ABORT BUSY ABORT NO\sCARRIER ABORT NO\sDIAL\sTONE ABORT
NO\sDIALTONE " " \pAT\r\c OK\r~2-\r\c-OK\r~2 ATdt\T\r\c CONNECT~90

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